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1 Introduction

2 Historical Trace of Capacity Utilization Measurement throughout Fordist to Post-Fordist era

This section reconstructs a historical trace of capacity utilization measurement not as a narrow sequence of statistical refinements, but as a history of state-crafting of capitalist development adopting operational routines of macroeconomic measurement within and across different institutional configurations looking at the case of the United States. Under a regulationist approach, here is argued that during the Fordist era, macroeconomic measurement of capacity utilization as a practice learned to make slack legible, actionable, and contestable. Its central premise is that operational routines are not neutral tools. Rather, they are institutionalized forms of ideological embeddedness shaping the institutional capacities to implement macroeconomic policy as *spatio-temporal fixes* by means of strategic selectivities on what counts as the relevant signal, which institutions are authorized to act on it, and over what temporal horizon they are expected to intervene. For that same reason, however, they can also become sources of dysfunction when the contradictions they once regularized deepen or shift. What matters, then, is not measurement practices themselves, but the political economy through which they are constructed, interpreted, stabilized, repurposed, and contested across changing accumulation regimes. In Jessop (2003)'s terms, these routines belong to spatio-temporal fixes that secure only provisional coherence and begin to unravel when the ideological commitments in which they are embedded are no longer capable of providing institutional coherence by means of displacing sources of stagnation and crisis junctures to other geographies or future temporal horizons (Jessop, 2006).

This perspective an historical anchor preceding the rise of American Keynesianism, departing by the earliest large-scale attempt to measure the slack of productive capacities in the United States. Interestingly, these efforts did not first emerge from a mature macroeconomic policy apparatus, but from the research program at the Brookings Institution, a long-standing policy think-tank within the landscape of private institutions shaping the *"policy"* debate in Washington, DC. In the aftermath of the Great Depression during the mid-1930s, Brookings' economists expanded empirical economic inquiry studying the production, consumption, capital formation, and income, at national scale, presenting the first empirical efforts to estimate the nation's productive ceiling sector-by-sector. By doing so, the debate and engagement with macroeconomic policy gain paramount self-reflexive capacities developing critical diagnosis identifying the very source of the great depression, *idle capacities*, into a measurable ceiling problem (Nourse et al., 1934). That shift pioneered later efforts to establish a policy dashboard of a broader set of operational routines. Long before American Keynesian became mainstream, *"America's Capacity to Produce"* – and its companions on issue of consumption among others, by the Brookings Institution made visible what counted as idle capacities in the first place. Capacity utilization should therefore not be approached as the progressive discovery of a timeless measurement artifact. It was assembled historically through institutional inquiry, statistical construction, and policy need. From the outset, the measurement issue was inseparable from the problem of governability.

The section follows that object across two historically distinct regime settings. Under Fordism, survey-based utilization measures were stabilized within a broader settlement in which American Keynesianism, business-information routines, and official statistical authority made slack governable for demand management. The crisis of the 1970s did not abolish those routines, but recoded them in a process that unfolded as a gradual institutional change (Mahoney and Thelen, 2009). Measures that had helped organize expansionary intervention became increasingly

read as warnings of inflationary pressures, while not here demonstrated, an interesting feature of this debate in historical perspective, is the participation of some of the main characters of neoliberal policies in the process of recodification from measuring capacity utilization as a slack index to an inflation sentinel as for example Alan Greenspan.

The measurement of capacity utilization during the post-fordist era can be interpreted as a discontinuity of the operational routines conducting monetary and fiscal policy, which became clear only ex-post it happened. Specifically in the aftermath of the so-called *"Volcker shock"*, amidst the rise of econometrics in the field of economics. Particularly, with the rising traction of time series econometrics applications to macroeconomics within the discipline. Jessop (2002) frames this shift as a reorganization of the forms through which accumulation was regularized: where Atlantic Fordism gave pride of place to wage and money forms institutionalized in the Keynesian National Welfare State (KNWS), the post-Fordist process of capitalist re-structuring which included the dismantling of the KNWS amidst new institutional forms of competition and correspondingly new ideological commitments the imperative functions of the state. Vidal (2013) places that reorganization in its historical setting by linking the crisis of Fordism to declining profitability, intensified competition, overaccumulation, and the rise of finance. In that setting, is the recodification of slack into a portable statistical device re-conceptualized from capacity utilization to output-gap from potential-output aligned with an anti-inflationary form of governance.

This shift also clarifies the stakes for the debates around business cycles and capacity utilization for researchers in political economy. Under a theoretical commitment of demand-led growth and distributive conflict, once money is understood as endogenous, the rate of interest can no longer be read as a market-clearing price but as a variable of distributive conflict where low-inflation is favourable for the banking-industry (Rochon and Setterfield, 2008; Lofaro et al., 2025). The institutionalization of interest-rates and inflation targeting as a policy variable set by central banks, then can be understood as an historically specific objective: inflation control while narrowing the space for contested forms of macroeconomic management. This is an essential axis on the rise of finance capital as the hegemonic force within the historical block of the transnational capitalist class (Robinson and Harris, 2000).

The stakes are not merely alternative traditions' preference for different measures. Capacity utilization measurements carry distinctive ideological commitments institutionalized in operational routines and bureaucratic practices, presented as uncontestable by means of their depoliticization (Best, 2025). Their construction, sedimentation, and recodification across accumulation regimes is the object of the historical trace that follows.

2.1 From Survey Measures to an Inflation Sentinel: Capacity Utilization Measurements during the Fordist Era

The Fordist history of capacity utilization measurement begins with a problem of governability rather than with a settled variable. In the United States, the Great Depression made idle resources visible as a political and administrative problem, but not yet as a stabilized object of macroeconomic management. What had to be built first was a way of rendering slack statistically legible. Brookings-era research was decisive because it turned unused productive capacity from an impressionistic sign of downturn into a measurable ceiling problem (Nourse et al., 1934). The issue was no longer simply whether factories stood idle, but how far actual output fell short of a plausible productive limit and under what conventions that limit should be defined. Capacity utilization thus entered macroeconomic discourse as an operational problem of visibility: institutions had to decide what counted as "capacity," how it could be approximated, and how that approximation could orient intervention. Later assessments made clear that this

early construction of the object rested on unresolved tensions between engineering and economic notions of capacity (Burns, 1954; Klein and David, 1958). The early history of utilization measurement was therefore already a history of governance by numbers rather than a neutral exercise in statistical description.

American Keynesianism gave that problem a clearer institutional and theoretical function. Hansen's interwar and wartime writings help explain why slack could no longer remain a residual category once stabilization became a policy objective (Hansen, 1936, 1941). Hicks and Samuelson then carried that problem into a more formal and pedagogical macroeconomic grammar, helping turn slack into a governable macroeconomic object rather than a residual business-cycle symptom (Hicks, 1937; Samuelson, 1947, 1948). Once the state was expected to manage demand, sustain employment, and monitor bottlenecks, it needed indicators that translated underutilized productive resources into signals for fiscal and monetary action. Capacity utilization became necessary not because Keynesianism discovered a timeless economic magnitude, but because a full-employment policy regime required an index through which effective-demand deficiencies could be rendered legible to authorities charged with stabilization. That shift mattered because it moved utilization measurement from descriptive inquiry into the core of a policy dashboard. The state did not govern an abstract "economy" in general. It governed through a portfolio of indicators that identified idle resources, pressures, and thresholds for intervention. In that setting, utilization was never merely technical. It was bound to a historically specific settlement in which demand management, employment commitments, and macroeconomic expertise reinforced one another (Barber, 1987).

That theoretical consolidation did not by itself settle the routine. McGraw–Hill did much of that work. Its plant-and-equipment surveys regularized managerial reports of actual and "preferred" operating rates and, in doing so, converted heterogeneous notions of productive capacity into a repeated business-information device (Butler, 1958; Phillips, 1963). What mattered here was less the elimination of definitional ambiguity than the creation of an administratively reproducible benchmark. Under Fordism, reproducibility itself became a source of authority. Repeated survey collection generated comparability across time and sectors, while the broader information infrastructure of monopoly capital helped stabilize the measure's legitimacy inside business forecasting and policy discussion (Baran and Sweezy, 1988; Munroe, 2007). What emerged, then, was not simply a better estimate of productive ceilings, but a routinized operational device through which firms, analysts, and policymakers could speak a common language about slack. At the same time, the benchmark's apparent neutrality rested on a dense set of conventions: assumptions about maintenance, overtime, sustainable operation, and the reporting practices of firms. Survey settlement therefore did not abolish mediation. It normalized it.

The Federal Reserve then transformed that settled routine into official macroeconomic authority. Once capacity and utilization measures were incorporated into the Industrial Production program and disseminated through the G.17 release, the survey benchmark ceased to be merely one possible way of gauging slack. It became the official routine through which the state rendered productive underuse visible (Board of Governors of the Federal Reserve System, 2017). That move altered the structure of the field. Institutional embedding did not simply add prestige to an existing measure; it raised the switching costs of alternatives and standardized what counted as admissible evidence. Physical-intensity measures such as Electric Motors Utilization and the Average Workweek of Capital remained conceptually important and in some respects tracked operational realities more directly, but they failed to secure comparable institutional adoption (Foss, 1963; Taubman and Gottschalk, 1971). The point is not that every rival was empirically defeated. It is that one routine proved easier to reproduce administratively, easier to integrate into policy practice, and easier to stabilize inside the state's statistical machinery. In this way, capacity utilization became an official governance variable of Fordist macroeconomic

management (Jessop, 2003).

The crisis of the 1970s did not abolish that routine. It recoded it. Under stagflationary pressure, capacity utilization was increasingly detached from its earlier role as a signal of effective-demand slack and reinterpreted as a warning device for inflationary strain. The benchmark survived, but the admissible meaning of the benchmark narrowed. What had once helped justify expansionary intervention now mattered insofar as it signaled when output was approaching inflationary constraint. This shift appears clearly in the Brookings debates, where capacity measures were increasingly judged by whether they could register inflationary pressure credibly (Klein et al., 1973). This was not a minor technical refinement. It marked a crisis-driven transformation in the policy dashboard itself. As the Bretton Woods order unraveled, the Fordist compromise weakened, and profitability pressures intensified, the utilization index was repurposed within a broader reorientation of macroeconomic governance. The same operational routine that had once helped organize full-employment management was now made compatible with emerging anti-inflation discipline. In regulationist terms, the benchmark persisted across a transition between modes of regularization, but its place in the hierarchy of policy priorities changed decisively (Aglietta, 2000; Jessop, 2002). That is why the recoding of capacity utilization should be read as a critical juncture in the fall of American Keynesianism rather than as a simple update in measurement technique.

What survived this transition was not Keynesian slack governance itself, but the institutional durability of the routine. Survey-based utilization retained authority because it had already been stabilized inside the policy apparatus, and that durability made repurposing easier than replacement. The Fordist history of capacity utilization measurement is therefore not a story of rise and disappearance, but of construction, routinization, recoding, and transition. Brookings made slack visible as a measurable problem. American Keynesianism turned that problem into a functional requirement of stabilization. McGraw–Hill and the Federal Reserve transformed the measure into a durable operational routine. Stagflationary crisis then narrowed its meaning into an inflation sentinel. That sequence matters for what follows. Once the Fordist benchmark had been recoded in this way, the ground was prepared for a new measurement regime in which slack would increasingly be governed not through survey-centered utilization series, but through more portable output-gap and potential-output benchmarks aligned with post-Fordist forms of discipline (Mahoney and Thelen, 2009).

2.2 The Rise of Econometrics and its Global Sedimentation on Institutions: From Capacity Utilization to Output-Gap Measurement through Post-Fordist Times

Post-Fordism, in Jessop's regulationist formulation, names not a periodization label but a reorganization of governance space under conditions of spatiotemporal compression—the acceleration of capital circulation, the rescaling of state authority toward supranational venues, and the displacement of nationally bounded Keynesian routines by portable benchmark architectures legible to transnational expert fractions and technopol networks (Jessop, 2002, 2004; Sklair, 2000). The governance demand this compression creates is historically specific: inherited Fordist routines—national surveys, dashboard indicators, discretionary fiscal instruments—cannot travel across the jurisdictions and institutional scales that globalized accumulation now requires. What emerges in response is a first tendency within post-Fordist macro governance: the withdrawal of the state from real-time discretionary management of demand and its replacement by rule-based, pre-announced policy frameworks whose authority rests not on correct tracking of the cycle but on the institutional consensus that depoliticized benchmarks can substitute for judgment (Jessop, 2002; Saad-Filho, 2018). Within this first tendency, the output-gap apparatus operates as a closure device—a portable, institutionally reproducible governance

variable that converts the unruly complexity of slack into a benchmark legible to surveillance bodies, budget offices, and central banks simultaneously. It does not track the cycle so much as close the governance problem that spatiotemporal compression has opened. Yet closure devices of this kind are themselves spatio-temporal fixes in Jessop's sense: they defer rather than resolve the contradictions of governance under globalized accumulation, displacing stagnation risks and crisis pressures across geographies and temporal horizons—until the conditions underwriting the fix shift, and the apparatus must repair itself or stretch beyond its original institutional design (Jessop, 2004).

The decisive critical juncture through which this first tendency consolidates is the Volcker Shock, but it should not be narrated as a simple victory of anti-inflation orthodoxy over Keynesian profligacy. Konings (2007) demonstrates that the institutional foundations of U.S. structural power in global finance were already being reworked through the re-emergence of international capital markets from the 1960s onward, and that the monetarist turn is better read as a reconfiguration of the relationship between U.S. state power, American finance, and the international monetary order than as mere submission to external discipline. Krippner's long-run definition of financialization—accumulation increasingly organized through financial channels rather than commodity production—gives that pivot its structural backdrop (Krippner, 2005), while Krippner (2007) traces how central bank transparency itself became a governance technology within the neoliberal order. Morgan and Sheehan (2015) serve as a historiographic corrective against reducing turning points to abstract theory disputes: elite narratives of reform characteristically present the reorganization of institutional power as technical correction, not as a political settlement with distributional consequences. Read together, these accounts make the Volcker moment legible not as a monetarist experiment but as the historical hinge through which a new benchmark regime—one premised on the credibility of depoliticized rules rather than the discretion of policy authorities—became institutionally plausible.

The theoretical legitimation for this regime shift had been constructed in the preceding decade through the Lucas critique, which Johnson (2024) reconstructs not as a laissez-faire argument but as an epistemological claim about the conditions of possibility for scientific governance. Lucas's program was premised precisely on the hope that states and scholars could make the economy visible and controllable—but only through stable, pre-announced policy rules that eliminated the Knightian uncertainty discretionary intervention produces. Discretionary policy, by making future governance inscrutable to agents, destroyed the rational expectations on which econometric legibility depended; it did not merely fail to stabilize but actively undermined the epistemic infrastructure that would have made stabilization scientifically defensible. The Volcker shock operates within this emerging disciplinary consensus as a discretionary act whose purpose is to establish the credibility that formal rules would later institutionalize—not a rule-based intervention in itself, but the violent founding moment of a regime that would subsequently present itself as rule-governed. What the 1980s achieved through institutionalized credibility among monetary policy agents, the 1990s would formalize as portable governance architecture: the Taylor rule, inflation targeting, and central bank independence statutes converted discretionary consensus into a framework defensible as technically neutral (Johnson, 2024; Taylor, 1993).

What the Volcker turn ultimately settles is a durable monetary-policy package that Saad-Filho (2018) names the New Monetary Policy Consensus. In his formulation, neoliberal monetary policy does not merely supply a superior stabilization technique; it constructs a political regime around inflation targeting, central bank independence, and floating exchange rates—a regime presented as best practice since the late 1980s, whose distributional consequences are rendered invisible precisely because the framework claims technical neutrality. Lofaro et al. (2025) reinforce this diagnosis from a Post-Keynesian and Sraffian standpoint, showing that the NMPC's governance of interest rates and inflation targets carries distributive effects that the consensus

framework systematically externalizes. The package matters for this section because it supplies the closure language through which slack is recoded: no longer principally idle capacity to be mobilized through demand management, but an output gap to be monitored, constrained, and interpreted through inflation expectations and nominal credibility. In other words, the post-Volcker settlement converts macroeconomic measurement from a demand-management lever into a rule-consistent benchmark apparatus—one designed not to track the cycle in real time but to replace real-time discretion with a rule that could be defended as technically neutral across institutional venues. That is the T1 formalization: the governance problem opened by spatiotemporal compression is closed not by better measurement but by measurement recoded as rule.

This settlement becomes durable because it is embedded in a comparative institutional spine rather than resting on one central bank model alone. The apparatus sediments across four privileged venues: the IMF, where potential output and the output gap operate as surveillance variables, fiscal-position benchmarks, and policy-recommendation inputs; the OECD, where they serve as medium-term, inflation-consistent benchmarks for Economic Outlook work and country reviews; the CBO, where they are folded into budget baselines under explicit medium-term closure assumptions; and ECFIN, where they are embedded directly in structural-balance calculations and rule-based fiscal governance through the Stability and Growth Pact. This is the moment when the output gap ceases to be one indicator among many and becomes a family of routinized devices—differentiated by institutional function but unified by a common closure grammar that recodes slack as an inflation-risk variable amenable to rule-consistent governance. The post-Fordist rise of the apparatus is therefore a story not of theoretical convergence alone but of institutional standardization across distinct policy venues, each of which gives the same benchmark a different governance lever while preserving its closure architecture. The two focal cases of this dissertation—the United States and Chile—represent structurally distinct articulations of this benchmark family: in the United States, the apparatus is most tightly embedded in budget-baseline interpretation and policy-rule discourse under financialized post-Fordism; in Chile, the same benchmark arrives through open-economy transmission logic and structural-balance governance, shaped by external constraint and the earliest concentrated experience of neoliberal restructuring in the periphery. Together, the two cases make legible a controlled comparison of how the same measurement apparatus produces different governance effects across core and peripheral institutional settings.

Within that spine, the IMF and OECD occupy the supranational benchmark tier. At the IMF, potential output and the output gap function as surveillance variables, fiscal-position benchmarks, and policy-recommendation inputs—a triple role that makes the device load-bearing across Article IV consultations, fiscal sustainability assessments, and program conditionality. The post-GFC methodological lane has added a layer of self-awareness: revision audits, uncertainty quantification, country-specific customization, hysteresis recognition, symmetry-bias diagnostics, and warnings about real-time misidentification now accompany what was previously an unqualified point estimate (Barkema et al., 2020; Cerra and Saxena, 2017; Cerra et al., 2023; Aiyar and Voigts, 2024). At the OECD, the device is framed differently—as a medium-term inflation-consistent benchmark embedded in Economic Outlook projections, country surveys, structural-balance calculations, and long-run scenarios—with post-crisis work emphasizing revision control and anti-cyclicality rather than abandonment of the estimator (Torres and Martin, 1990). The result is revealing: post-GFC reflexivity does not displace the output gap from the apparatus but repackages it as a more cautious and better-calibrated benchmark, one whose survival depends on absorbing critique rather than resolving it. That pattern—reflexive adaptation without paradigm exit—is one of the clearest signatures of how the apparatus reproduces itself under stress.

The CBO and ECFIN reveal a different and more coercive embedding of the same benchmark

family. In the U.S. case, potential GDP and the output gap are tied directly to the budget baseline and to standardized interpretations of the budget balance under explicit medium-term closure assumptions—an architecture that makes the CBO's output-gap estimate consequential not because it is epistemically superior but because it is administratively binding (Congressional Budget Office, 2004). In the EU case, potential growth and output gaps are folded into rule-based governance through the Stability and Growth Pact, the European Semester, and structural-balance calculations, where they adjudicate what counts as fiscal space, discipline, or excess (Havik et al., 2014). Here the output gap is no longer only a surveillance device; it becomes part of the machinery through which fiscal sovereignty is constrained by a measurement routine that presents itself as technically neutral. The same benchmark family that in one institutional venue appears as assessment appears in another as enforcement—and it is precisely this functional versatility that makes the apparatus durable across governance settings that differ in everything except their reliance on the output gap as a closure variable.

The U.S. focal case condenses this post-Fordist settlement with particular clarity. The technical map identifies the U.S. lane as combining CBO closure logic, IMF methodology repair, policy-rule embedding through the Taylor tradition, contemporary filter sensitivity, and a retained Blanchard–Quah structural-decomposition branch in the background (Congressional Budget Office, 2004; Adams and Coe, 1990; Taylor, 1993; Alichi, 2015). Its defining asymmetry is that the output gap is especially tightly linked to budget-baseline interpretation and policy-rule discourse—a linkage that makes the benchmark consequential not only for monetary but for fiscal governance under financialized conditions. That makes the U.S. case useful not because it is universal but because it reveals the strongest articulation between financialized macro governance, benchmark authority, and fiscal interpretation. **?** sharpens this point by demonstrating that filter-based output-gap estimates in the U.S. policy-rule context exhibit substantial real-time instability—revisions large enough to alter the sign of the gap and, with it, the policy signal the benchmark is supposed to deliver. In this configuration, the output gap does not merely describe slack; it organizes how slack becomes administratively intelligible for macroeconomic management, even as the real-time reliability of that organization remains structurally precarious.

Chile, by contrast, shows how the same apparatus family is articulated differently under open-small-economy conditions shaped by external constraint, commodity dependence, and the earliest concentrated experience of neoliberal restructuring in the periphery. The focal map identifies the Chile lane as combining the IMF apparatus core, BCCh apparatus core, BCCh real-time and revision governance, institutional doctrine from BCCh, Hacienda, and Dipres, and an IMF open-small-economy transmission companion (Magud and Medina, 2011; Fuentes et al., 2008; Banco Central de Chile, 2017; Ministerio de Hacienda de Chile, 2023; Dirección de Presupuestos, 2025). Its defining asymmetry—stronger structural-balance embedding and open-economy transmission logic—gives the benchmark a different governance lever than in the U.S. case: here the output gap mediates not primarily budget-baseline interpretation but the relationship between commodity-cycle exposure, inflation pass-through, and fiscal-rule compliance. Figueroa et al. (2019) make visible a further layer of this embedding by documenting how output-gap revisions in the Chilean context propagate through inflation forecasts and fiscal-balance calculations—a revision politics in which the benchmark's instability is not merely epistemic but has direct fiscal consequences. Chile thus operates as the clearest comparative reminder that portability does not imply homogeneity: the benchmark survives across institutional settings, but its governance leverage depends on the macro-structural and institutional terrain in which it is embedded.

The crisis of 2008 does not dislodge this settlement; it exposes the conditions on which the first tendency's viability has always depended. Best (2025) reconstructs the political economy of monetary depoliticization in the UK to show that rule-based governance frameworks hold only while the narrow technical devices on which they rest retain credibility—and that credibility

requires a historically specific set of macroeconomic conditions: moderate shocks, anchored expectations, and institutional environments where pre-announced targets can absorb short-run deviations without triggering the political backlash depoliticization was designed to foreclose. Post-2008—and more sharply after the COVID disruption—these conditions erode, and a second tendency becomes visible: the statistical and information infrastructure assembled during three decades of benchmark governance now makes real-time intervention technically feasible in ways that were unavailable to Fordist or early post-Fordist states (Foster, 2022). Central banks do not merely repair the old rule-based architecture; they begin to reoccupy the discretionary governance space that the first tendency had explicitly vacated—expanding mandates, deploying unconventional instruments, and interpreting their targets with a flexibility that the original depoliticized framework was built to prohibit. The filtering lane reaches a parallel conclusion from the methodological side: output-gap estimation rests on competing devices with different closure assumptions, revision properties, and crisis sensitivities, and there is no single technical winner—only a portfolio of tools whose coexistence is itself evidence that the benchmark can no longer be defended as a settled, rule-consistent variable (Hamilton, 2018; Quast and Wolters, 2022; Durand and Fornero, 2024). The post-GFC period should therefore be read not as collapse but as the moment when T1 institutional architecture begins to bear T2 operational demands—a mismatch whose sharpest expression is not methodological plurality but the legitimacy crisis it produces.

The ECB provides the sharpest meso-institutional illustration of this T1/T2 mismatch. Argyroulis and Vagdoutis (2025) show that the ECB's early strategy statements were built for a T1 world: tightly defined performance criteria, a narrow price-stability mandate, and an institutional self-understanding premised on the claim that technocratic governance required neither democratic authorization nor discretionary judgment beyond the application of pre-announced rules. The 2021 strategy revision installs a different operational reality. It broadens the framework, expands discretionary space, incorporates climate and distributional considerations, and reinterprets the mandate in ways that Argyroulis and Vagdoutis characterize as rendering it "broad and vague rather than clear and narrow" (p. 14)—a formulation that captures precisely the institutional diagnosis at stake. What they name is not a failure of implementation but a structural contradiction: the apparatus must stretch its discretion to absorb post-crisis demands—multiple objectives, flexible medium-term orientation, unconventional instruments—yet in doing so it dismantles the very claim to neutral, rule-bound expertise on which its depoliticized authority was constructed. The result is not the end of technocratic governance but a structurally weakened version of it—one that must now justify its discretion rather than simply assert its technical neutrality, and that can no longer present the output gap as a self-evidently rule-consistent variable. This is not a contingent difficulty at the edge of the system; it is the concentrated institutional expression of a regime whose first-tendency architecture was not designed to carry the governance demands its own crisis management now requires.

Once seen from this angle, methodological plurality is not a technical side note but a political symptom of an apparatus repairing itself for second-tendency demands using first-tendency tools. The filtering branch—Beveridge–Nelson decomposition, Hodrick–Prescott, band-pass variants, Hamilton-type structural critiques, and the later repair literature—does not converge on one universally superior output-gap estimator; these devices coexist as a portfolio whose usefulness depends on data-generating structure, intended policy use, institutional benchmark requirements, and tolerance for revisions (Beveridge and Nelson, 1981; Hodrick and Prescott, 1997; Baxter and King, 1999; Christiano and Fitzgerald, 2003; Hamilton, 2018; Franke et al., 2024; Siemers, 2025). No single technique wins because the governance problem the apparatus must solve has shifted: what was once a question of selecting the correct filter for a rule-consistent benchmark is now a question of which combination of devices can sustain benchmark authority under conditions where the rules themselves are being renegotiated. The technical

branch is therefore saturated for the purposes of §2.2—not because one method has been vindicated, but because the apparatus survives by combining and repurposing coexisting devices while preserving the output gap as the dominant governance variable. Plurality is contained inside the benchmark regime rather than opening it to paradigmatic challenge.

The payoff for the rest of the chapter is now legible. Post-Fordist macro governance globalized the measurement of slack by embedding the output gap in surveillance, budgeting, structural-balance rules, and central-bank strategy statements, while financialization and technocratic statecraft made this benchmark increasingly authoritative—portable enough to travel from Washington to Santiago, from Article IV consultations to structural-balance calculations, from Taylor-rule calibrations to ECB mandate revisions. But the very durability of the apparatus exposed its limits: it is revision-prone, closure-dependent, and institutionally selective, and its post-crisis survival increasingly depends on the discretionary repair and broadened mandates that its original depoliticized architecture was designed to render unnecessary. That is why the heterodox reopening of capacity utilization in the next section becomes historically intelligible. It does not arise in an empty theoretical field; it emerges against the dominance of an output-gap apparatus that has become globally portable precisely by turning one recoded form of slack into a governing norm—and against the growing visibility of the contradictions that portability produces when the governance demands outrun the institutional design.

2.3 The Capacity Utilization Controversies in Political Economy Debates

The debate over the measurement of productive capacity and potential output is not confined to technical economics. When estimates of the output gap serve as the operative benchmark for structural-balance fiscal rules (as they did across the Eurozone in the aftermath of the 2008 crisis), their systematic bias carries observable distributional consequences. Klär (2013) documents how unreliable potential-output and NAIRU estimates constrained fiscal policy in Europe, producing contractionary impulses in economies operating well below productive capacity. Heimberger and Kapeller (2017) show how the European Commission's model for estimating 'potential output' (Havik et al., 2014), are used to calculate 'structural budget balances', which in turn translate into country-specific structural-balance governance reinforcing pro-cyclical austerity logic. Heimberger (2020) tracks how European Commission potential-output revisions reshaped fiscal space under crisis conditions, compressing public investment precisely when demand shortfalls were largest. From a classical-keynesian perspective, Palumbo (2015) provides a theoretical restatement of Okun's Law explaining what these findings imply: potential output is endogenous and path-dependent rather than an exogenous supply ceiling. Hence, the measurement apparatus was not merely imprecise but conceptually misconceived as a governance instrument. Mismeasurement, in this register, is not a technical nuisance—it allocates the costs of under-utilization to workers and the public sectors while insulating financial creditors from adjustment. That the dominant measurement instrument produces such consequences with such regularity raises an immediate question for political economy as a research program: if the dominant measurement instrument is consequentially wrong, what alternative does the tradition offer?

The answer reveals a problem unresolved within Political Economy debates, which expresses itself with the standard practice of research to engage capacity utilization taking the Federal Reserve Board series as an empirical object taken at face value without further scrutiny on its methodological construction. As showcased above, the FRB capacity utilization series was not produced in a vacuum, but produced as a measurement of governance under a historically contingent institutionalization where: monitoring manufacturing slack for monetary policy purposes within the institutional landscape described in 2.1. Its migration into political economy research happened without systematic evaluation of its adequacy for subjecting theoretical claims

to empirical scrutiny. The series existed, covered the relevant variables, carried official authority, and offered long historical coverage. These are properties that make an instrument useful for governance dashboards. They are not properties that make it adequate for adjudicating theoretical claims about the long-run dynamics of capitalist accumulation; no such distinction was drawn at the point of adoption. The instrument was recruited from a governance context into a scientific one without the transit being examined.

The historical timing of this migration compounds the problem. The political economy debate over the normal rate of capacity utilization gains its contemporary momentum in the aftermath of the Global Financial Recession in which Political Economy approaches to macroeconomics have gained traction amidst the crisis of the Post-Fordist regime. As portrayed in 2.2, the output-gap measurements constituted themselves as an statistical apparatus embedded across supranational institutions building the operational infrastructures of fiscal surveillance. The same class of instrument is simultaneously institutionalized as governance infrastructure by mainstream bodies and recruited as scientific evidence by political economy researchers pursuing opposed theoretical agendas. A further scope constraint operates silently throughout: FRB is an industrial manufacturing survey. In the Fordist era this covered a dominant sector of accumulation. In the Post-Fordist economy it traces a contracting share of output, employment, and investment while being used to characterize economy-wide capacity dynamics. The representational problem deepens across the period of the instrument's heaviest use in political economy research.

The controversy over capacity utilization's long-run behavior carries substantive theoretical stakes that deserve clear statement before their empirical impasse is diagnosed. The Neo-Kaleckian tradition argues that the normal rate of capacity utilization is endogenous to demand. In this view, firms revise their desired utilization in response to persistent demand conditions: through gradual adjustment of expectations (Lavoie, 1995), through micro-founded shift-choice mechanisms in which indivisibility makes utilization respond when capital cannot be smoothly resized (Nikiforos, 2013), or through hysteresis in which the history of demand shocks durably reshapes the utilization target itself (Setterfield and David Avritzer, 2020). The implication is that actual utilization need not converge to any structurally given benchmark: demand conditions shape long-run accumulation trajectories and distributive outcomes are path-dependent. The Sraffian-Keynesian tradition advances the opposed claim. In the classical formulation, the normal rate is determined by cost structures and competitive conditions, and actual utilization gravitates toward it over the long run, while the speed of convergence might be mediated due sluggish investment and a desired rate with slack to uncertainity of demand surges (Ciccone, 1986).

The contemporary version is more precise: within the Sraffian supermultiplier framework, autonomous demand drives the growth trend of output while investment adjusts to bring capacity in line with demand, so that utilization converges toward a normal rate determined by structural conditions rather than by demand history (Girardi and Pariboni, 2016). The convergence claim is buttressed by cross-country evidence: Gahn and González (2022) document stationarity of utilization across twenty-one countries and find positive short-run but negligible long-run correlation between utilization and growth—a pattern they read as more compatible with convergence than with permanent Neo-Kaleckian drift.

If utilization gravitates to a structurally given normal rate, then demand-side interventions have only transitory effects on accumulation—the system self-corrects through investment adjustment. If utilization is path-dependent, then persistent demand shortfalls generate durable consequences for both investment and employment, which means macroeconomic governance can reshape accumulation trajectories rather than merely smoothing cycles. Resolving the controversy empirically requires a measurement instrument whose properties carry genuine in-

formation about the long-run relationship between actual and normal utilization.

Nevertheless, that requirement has not been met. Both sides of the controversy require a measure whose stochastic properties (stationary or trending, mean-reverting or drifting) carry information about whether actual utilization converges to a normal rate over the long run. The Federal Reserve Board series cannot provide this, because its statistical behavior is not a property of the underlying economic process but an artifact of construction. Capacity Utilization series built by the FRB combines survey responses on current operating rates with regression-based capacity estimates that are smoothed against capital stock and time trends, explicitly adjusted to move the resulting series away from an engineering concept of maximum output toward an economic concept of sustainable output under normal operating conditions (Shapiro et al., 1989). The stated design objective is that the resulting measure has as a byproduct an engineering mean-reversion behavior by construction, and is impeded of registring evidence of chronic departure from normal capacity utilization:

Testing whether the FRB series is stationary is therefore not a test of whether the actual utilization rate converges to a normal rate fixed by cost conditions; it is a test of whether the Federal Reserve's construction procedure successfully implemented its own design objective. Nikiforos (2016, 2021) provide the most systematic articulation of this problem, arguing that FRB is closer to a cyclical slack indicator than to a measure of long-run movements in normal utilization. The relevant time horizon for the controversy—on the order of the useful life of capital, roughly thirty years—far exceeds the frequency at which the series is informative.

The construction opacity compounds the difficulty. The raw micro-data underlying the series is not publicly available, the weighting procedures combining different survey sources are not fully disclosed, and the splicing across institutional transitions lacks transparent documentation (Board of Governors of the Federal Reserve System, 2017). The instrument cannot be subjected to independent methodological scrutiny by the researchers who rely on it. The category error is therefore precise: FRB was built to produce a stable, institutionally credible signal for monetary policy governance—stability is a virtue for that purpose and a disqualifying defect for adjudicating a controversy that turns on whether utilization is stable or trending. But even if FRB were replaced by a transparent and well-constructed survey, a deeper problem would remain, one rooted not in the instrument's opacity but in the micro-level premise that all survey-based utilization measures share.

Even if a transparent and well-constructed survey replaced FRB, the deeper problem is that both the Neo-Kaleckian desired rate and the Sraffian normal rate presuppose that firms have a coherent utilization target, whether endogenously revised or exogenously anchored, that can in principle be elicited through survey instruments. This shared micro-level premise is empirically unsupported. The difficulty begins with scope: FRB covers manufacturing, which in the Fordist era traced a dominant share of accumulation but in the Post-Fordist economy represents a contracting fraction of output, employment, and investment (Haluska, 2023). The measurement object thus narrows as the theoretical ambition widens. But the problem runs deeper than sectoral coverage. Management and operations research literature converges on a portrait of capacity utilization as a neglected variable at firm level, managed tactically for short-run performance rather than optimized toward a long-run target rate (Anderson, 2001; Singh et al., 2022). Planned utilization levels are simultaneously conditioned by competitive pressures, demand expectations, and cost structures—outcomes of contradictory organizational logics rather than expressions of a unified firm objective (Lieberman, 1989).

The structural source of this incoherence is the transformation of corporate governance itself. Vidal (2022) provides the most pointed formulation: the Post-Fordist firm is not a unified decision-making agent. Plant-level managers maximize throughput against cost-cutting constraints. Corporate executives manage earnings per share, share buybacks, and stock price

performance under the shareholder-value regime that restructured Anglo-American corporate governance from the 1980s onward (Lazonick and O'Sullivan, 2000). Directors mediate agency conflicts between management and ownership whose objectives are misaligned by design (Fligstein and Goldstein, 2022). Even if every tier responds honestly to a utilization survey, the aggregated response does not represent a coherent firm-level desired or normal rate—it represents the resultant of an internal contest whose outcome has no clean theoretical interpretation on either side of the controversy. The irony is temporal: shareholder-value capitalism decouples plant-level utilization decisions from corporate accumulation strategy during the period when the political economy measurement debate intensifies. The measurement object becomes less theoretically interpretable exactly when it is most heavily used. The works by Nikiforos recognizes this cluster of problems more clearly than any other participant in the debate, and his attempt to address it through measurement rescue is the most self-reflexive move within the tradition.

Nikiforos (2016) engages the measurement debate by subjecting the FRB series to systematic scrutiny on its own terms. His critique of construction opacity, respondent ambiguity, and engineered mean reversion is well-founded and independently motivated: the problems identified in the preceding paragraphs are substantially his diagnosis. His proposed rescue is correspondingly analogous: the average workweek of capital, developed by Foss (1981, 1985) among others, is a direct physical measure of machinery use, sidesteps both the construction artifacts of FRB and the micro-level premise problem, since it measures capital intensity rather than eliciting managerial assessments of a desired operating rate. Nikiforos (2021) deepens the case by arguing that AWW and related physical proxies are conceptually better matched to the long-run horizon relevant to the controversy. This is a genuine methodological advance. The question is whether it can settle the dispute. AWW trends where FRB does not, and the cointegration of FRB on AWW is read as evidence that FRB's stationarity reflects measurement error rather than true convergence. But this leaves open whether the proxy's non-stationarity is evidence for endogenous normal utilization or simply a property of a different measurement object—and the single-equation ARDL framework within which the evidence is produced cannot adjudicate that question from within.

Gahn and González (2020) press the empirical side of this concern, arguing that the ARDL evidence does not robustly establish that AWW delivers the stochastic-trend story the Neo-Kaleckian interpretation requires. Their critique is effective as a rebuttal of Nikiforos's specific econometric exercise. But it does not itself escape the same methodological terrain. The Sraffian-convergence response to the controversy relies on a set of stationarity tests, proxy admissibility arguments, and transitory-shock evidence (Gahn and González, 2022; Deleidi et al., 2024); yet demonstrating that the Neo-Kaleckian case is empirically insufficient is not the same as positively identifying the long-run adjustment mechanism through which utilization is supposed to converge. What the debate has established, then, is not which tradition is right but that the instruments available to either side cannot settle the question. Both sides of the controversy appear to remain confined to single-series proxy disputes and single-equation testing strategies that may not be well suited to recover the structural object at stake. If the long-run relationship between actual utilization and the capacity benchmark is shaped by the joint dynamics of accumulation, distribution, and technical change, it may be better understood as a system property than as a property of any single observable series. Unit root tests, proxy comparisons, and conditional error-correction models applied to individual series are not well suited to identify it. Whether the controversy is blocked by insufficient data or by the limits of the identification strategies available to it remains an open question—but one that the measurement debate in its current form does not appear equipped to resolve.

2.4 Shaikh's Critique and the Requirement for Structural Identification

Shaikh's interrogation of the measurement apparatus is the only intervention in the political economy literature that proceeds independently of the utilization controversy itself — not as a contribution to the Kaleckian–Sraffian debate but as a critical assessment of the data infrastructure underlying the rate of profit. The tradition from which it emerges begins with Foss (1963), who constructed the Electric Motor Utilization index by comparing actual electricity consumption in manufacturing with the theoretical maximum implied by installed motor horsepower, producing a physically grounded intensity measure that registered higher utilization rates and greater long-run variation than the Federal Reserve series. Shaikh (1992) extended the method, building a continuous EMU series from 1899 to 1963 using Census data and adapting Christensen and Jorgenson's (1969) estimates, then splicing it with McGraw-Hill survey data to construct a utilization measure covering 1947 to 1985; the methodology underlying this lineage is documented in Appendix C. Shaikh's critique of the FRB series operates on two levels.

At the institutional one, the Federal Reserve combines survey data through undisclosed procedures and adjusts the resulting estimates with the stated objective that the measure should not be chronically below normal, making stationarity a design feature rather than a discovered property of the data (Shapiro et al., 1989, pp. 187–188). At the conceptual level, physical measures reveal non-stationary, asymmetric fluctuations with greater amplitude than FRB, suggesting that the survey-based series suppresses precisely the long-run patterns that matter for the theoretical question. Nikiforos (2016) later formalized this critique through the plant-manager thought experiment: because adding or dropping shifts changes the definition of normal operating conditions, the survey denominator moves with the numerator, making the instrument self-referential with respect to the object it purports to measure.

Shaikh's proposed solution — recovering productive capacity as the long-run component of the output–capital cointegrating relation under a maintained normal-utilization closure — represents an advance over both survey-based measures and proxy comparisons; the method and its econometric implementation are developed in §**??**. The substantive empirical finding is unbalanced growth: the growth rate of productive capacity persistently diverges from the growth rate of capital accumulation, breaking the balanced-growth assumption shared implicitly by both the Kaleckian and Sraffian positions. The transformation elasticity θ linking capacity expansion to capital accumulation is not unity. Shaikh showed it is approximately 0.66 for the postwar US corporate sector — each percentage point of capital growth generates only two-thirds of a percentage point of capacity growth. The limit of the approach is that the single-equation specification recovers this elasticity as a fixed parameter, averaged over the entire sample, without a channel through which institutional variation can enter. Whether this fixity is a property of the object or an artifact of the identification strategy is a question the historical trace cannot answer. It requires a conceptual framework that defines the theoretical object the elasticity proxies for, and a replication discipline that tests whether the fixity survives systematic confinement.

The next section provides both. It derives the accounting laws of capital accumulation under unbalanced growth, identifies the transformation elasticity $\theta(\Lambda)$ as the regime-specific structural parameter governing the capital-to-capacity mapping, locates Shaikh's identification strategy within that accounting structure, and establishes the replication discipline through which the fixed-parameter assumption can be tested. The measurement controversies traced in this section — survey self-reference, filter-imposed stationarity, and the suppression of long-run patterns — define what an adequate identification must avoid. The conceptual framework that follows defines what it must recover.

3 Conceptual Framework

3.1 Reproduction, Smooth Reproduction, and the Say's Law Benchmark

The previous section traced the measurement of capacity utilization from its Fordist-era survey origins through its post-Fordist sedimentation into the econometric and filter-based apparatus that governs fiscal policy in the United States and Chile. That trace ended with a diagnostic: Shaikh's cointegration-based identification recovers unbalanced growth—a genuine advance—but he does not develop a substantive interpretation of the long-run parameter beyond the intuition that capacity utilization is the non-cointegrated component of capital accumulation. The strength of his methodology, on my reading, is precisely the identification of a long-run transformation elasticity of productive capacity to capital stock. Yet the implementation leaves that parameter poorly theorized: capital embeddedness and autonomous technological change are assumed rather than derived, and the stability of the fixed parameter is never tested—leaving open whether constancy is a structural feature of the data or an artifact of the ARDL bounds-testing procedure itself (Pesaran et al., 2001). Several theoretical commitments developed at length throughout *Capitalism*—endogenous technological change, gravitational instability — find no expression in this empirical specification (Shaikh, 2016).

Before the critical replication developed here proceeds, the object it targets must be derived. This section builds the accounting framework from which $\theta(\Lambda)$ emerges as a regime-specific property of capital accumulation under unbalanced growth, situates Shaikh's empirical strategy within that framework, and identifies the specific site where critical replication intervenes.

The accounting structure of capital accumulation does not begin with an econometric specification. It begins with a question Marx posed in Volume II of *Capital*: under what conditions can the capitalist system reproduce itself? The reproduction schemas formalize this question by decomposing the aggregate economy into departments whose output must satisfy both the value requirements of surplus extraction and the use-value requirements of material replacement. The circuit of capital—money capital advanced, productive capital employed, commodity capital realized—imposes temporal and proportional constraints: commodities must be sold, value must be realized through circulation, and the material composition of output must be reproduced in the right proportions across departments for accumulation to proceed (Marx, 2000).[1]

What the reproduction schemas establish is that production alone is insufficient for accumulation. Realization requires that the output of each department finds a market in the other departments' demand, and that the aggregate proportionalities hold across the circuit. When they do not—when disproportionality between departments exceeds the absorptive capacity of the market—the circuit breaks and crisis intervenes. The schemas therefore define the *conditions of possibility* for smooth reproduction, not its inevitability.

Shaikh (2016)'s engagement with this framework in *Capitalism* is tangential and, for the purposes of the present chapter, consequential in its absences. The reproduction schemas appear in his argument at three sites: as a device for demonstrating the endogeneity of business savings at the aggregate level, where the proportionality condition between departments is declared dispensable (Ch. 13, §III.2); as a price-theoretic benchmark identifying the output proportions under maximum expanded reproduction at which aggregate profit is invariant to changes in relative prices (Ch. 6, §IV.2); and as the source of the upper bound on accumulation, $g \leq r$, within a crisis theory organized entirely around the profit rate as the sole mediating variable (Ch. 13, §V). Each of these engagements instrumentalizes the schemas for a purpose other than reproduction analysis: the first dissolves the departmental structure into an aggregate identity, the second subordinates it to the transformation problem, and the third reduces it to a

[1] For a contemporary integrated reconstruction, see Basu (2022, Ch. 4).

boundary condition on growth.

The entire tradition that derived crisis theory from the reproduction schemas—the Luxemburg–Hilferding–Grossmann line—is absent from *Capitalism*. This is not an oversight but a logical consequence of Shaikh's analytical operations on the schemas: if the sectoral issue disappears at the aggregate level and *all* crisis dynamics pass through profitability, then there is no conceptual location where disproportionality as a distinct mechanism could be formulated. What is foreclosed across all three sites is precisely the distinctive analytical content of the schemas: interdepartmental proportionality as an independent structural constraint on accumulation—one that is not reducible to aggregate demand, relative prices, or the profit rate.

This foreclosure has a direct consequence for the object of the present chapter. If the departmental structure carries no independent analytical weight, the transformation elasticity θ has no theoretical reason to vary across accumulation regimes—it is just a technical parameter linking two aggregates within socially constructed economies understood as a single totality, traceable only at different analytical moments (Vidal and Peck, 2012). But if the reproduction schemas are taken seriously as an accounting framework, then θ not only represents persistent unbalanced growth in the long run but also carries the regime-dependence of the proportional interdepartmental conditions for smooth reproduction: the composition of investment across departments, the material requirements of expanded reproduction, and the institutional conditions under which those requirements are met or violated. The accounting laws derived in the following section identify why the notion of regime-dependence under unbalanced growth is critical for understanding the dynamics of capital accumulation. There, θ represents the transformation of capital stock into productive capacity, while Λ denotes its institutional configuration—hence the notation $\theta(\Lambda)$.

This chapter takes the reproduction schemas as the accounting foundation and works under a classical long-run closure in which all produced commodities are realized, all capacity is utilized, and the economy operates on a "golden path" of smooth reproduction (Basu, 2022). This assumption can be traced back to Say's law,[2] $\mu \equiv Y/Y^p = 1$: supply creates its own demand, markets clear continuously, and the economy works under full capacity utilization—or, in neoclassical jargon, the output gap vanishes. Under this methodological *closure*, the temporal and proportional constraints of the reproduction schemas are satisfied by assumption. The accounting identities of capital accumulation—the rate of accumulation $k = \phi^p B$, the investment share at normal capacity $\phi^p \equiv I/Y^p$, and capital productivity $B \equiv Y^p/K$—hold without the demand side.

Smooth reproduction is not a description of reality but a benchmark of its stringent conditions (Okishio, 2022). It is an analytical device that isolates the *supply-side accounting structure* of capital accumulation from the demand conditions governing whether the productive structure can be realized. By abstracting from $\mu \neq 1$, the tendential laws of accumulation can be derived at the level of accounting identities alone—before any investment function, Phillips curve, or demand-formation equation is imposed—providing the theoretical underpinnings for Shaikh's methodology. The cost of the abstraction is that the entire demand side of the accumulation equation is suppressed, with degenerate consequences that can only be understood once effective demand stabilization is recognized as the force that breaks them. The benefit is that the structural parameter governing the transformation of capital into productive capacity—$\theta(\Lambda)$—can be isolated and its consequences traced analytically, while Shaikh's own framework for estimating capacity utilization can be subjected to scrutiny and refinement.

[2] Which here, as argued by Foley (1985), is used as a mere abstraction.

3.2 Laws of Capital Accumulation under Unbalanced Growth

The cost of the smooth reproduction abstraction imposed in the previous section is not merely conceptual. It produces degenerate results at the boundary of the feasible interval. As the rate of capital accumulation approaches unity, productive capacity outruns realized output, the labor supply is exhausted, the consumption share collapses, and μ falls below 1. The three channels that re-enter simultaneously—labor market tightening, demand shortfall, and falling utilization—are the terrain of the class struggle over distribution, utilization, and the institutional forms that govern whether accumulation can proceed. That terrain is developed in Chapter 2. Here, the smooth reproduction benchmark ($\mu = 1$) is maintained in order to derive the accounting laws that define the object the empirical strategy must recover. The degeneracy is not a defect of the accounting structure but a signal of its scope: it marks the point where pre-behavioral laws exhaust themselves and demand conditions must be imposed.

3.2.1 The Rate of Capital Accumulation and the Three Channels of Distribution

The gross rate of capital accumulation decomposes as:

$$\hat{k} + \delta = \frac{I}{K} = \phi^p B \tag{1}$$

where $\phi^p \equiv I/Y^p$ is the investment share at normal capacity—the degree to which investment claims the productive capacity of the capital stock, independent of realized demand conditions—and $B \equiv Y^p/K$ is capital productivity at normal capacity.[3]

Distribution does not appear as an explicit argument, but it is not absent. It enters through three channels of unequal visibility: μ carries demand conditions shaped by the distributive conflict over utilization; B carries the indirect distributional mapping through $\theta(\Lambda)$, connecting capital productivity to the mechanization choice endogenous to the wage-profit relation; and ϕ^p carries the structural power of capital over the surplus—the capacity of capitalists to direct a share of normal-capacity output toward accumulation rather than consumption. The algebraic cancellation of μ—which drops out of $\phi^p B$ under $\mu = 1$—is real; the social cancellation is not.

3.2.2 The Unbalanced Growth Closure

The transformation elasticity of capital stock into productive capacity is defined as:

$$\theta(\Lambda) \equiv \left.\frac{\partial \ln Y^p}{\partial \ln K}\right|_{\Lambda} \tag{2}$$

It measures how efficiently accumulation generates productive capacity—whether each unit of capital deployed expands, maintains, or erodes Y^p relative to K. At a given institutional configuration Λ, θ is fixed: the regime determines the structural conditions under which mechanization choices are made, pinning the transformation elasticity as a parameter of the regime.

The methodological closure of the system is:

$$\hat{y}^p = \theta(\Lambda) \cdot \hat{k} \tag{3}$$

This is the unbalanced growth closure at the origin of the mechanization possibilities frontier. Balanced growth—$\theta(\Lambda) = 1$, capital and productive capacity expanding in exact proportion—is

[3] The gross investment rate I/K has no dedicated symbol; when needed it is written as $\hat{k} + \delta$ or equivalently as its decomposition $\phi^p B$. This convention prevents the gross/net conflation that can vitiate the derivation of the speed equation.

the knife-edge special case. The general case is $\theta(\Lambda) \neq 1$, and the tendential laws of over- and under-mechanization follow directly from which side of unity $\theta(\Lambda)$ sits on.

Two immediate consequences follow from the closure and the accounting identities. First, capital productivity growth:

$$\hat{b} = \hat{y}^p - \hat{k} = (\theta(\Lambda) - 1)\,\hat{k} \tag{4}$$

Capital productivity grows when $\theta > 1$ (under-mechanization: accumulation raises B), shrinks when $\theta < 1$ (over-mechanization: accumulation erodes B), and is constant at the knife-edge. This is a pre-behavioral result. Second, the dynamics of the investment share:

$$\hat{\phi}^p = \hat{\imath} - \hat{y}^p = \hat{\imath} - \theta\,\hat{k} \tag{5}$$

The growth rate of the investment share is the gap between investment growth and capacity growth. A critical feature of the accounting structure is that log-differentiating equation (1) yields:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta)(\hat{\imath} - \hat{k}) \tag{6}$$

an identity in which θ nets to zero. The closure enters through $\hat{b}$ and through $\hat{\phi}^p$ with opposite signs and cancels. The three equations—the accounting identity, the closure, and the productivity consequence—are underdetermined for the dynamics of $\hat{k}$: the system has three unknowns ($\hat{k}$, $\hat{y}^p$, $\hat{\imath}$) and two independent relations.

3.2.3 Investment-Savings Closure

The missing ingredient is a closure on investment. Under smooth reproduction ($\mu = 1$), the goods market clears: investment equals savings. Expressed in capacity shares, this equilibrium condition is:

$$\phi^p = s \qquad \text{(constant)} \tag{7}$$

The share of productive capacity claimed by investment is structurally fixed. This is the first behavioral assumption in the system. It closes the underdetermined accounting structure of the previous section by pinning the investment share, so that $\hat{\phi}^p = 0$ and investment growth is tied to capacity growth: $\hat{\imath} = \hat{y}^p = \theta\hat{k}$. The system closes and θ-dependent dynamics emerge.

The condition $I = S$ in capacity shares is coherent with the smooth reproduction benchmark already imposed, and its economic content varies across traditions of political economy. In Marx's reproduction schemas, the condition corresponds to the total reinvestment of surplus value: the surplus product is entirely directed toward expanded reproduction, and the proportionality conditions of the two-departmental schema ensure that the material composition of the surplus matches the input requirements of both departments (Marx, 2000). In the classical tradition, savings out of profits determines investment, and the profit rate adjusts through competition to enforce the condition. In the Keynesian tradition, the same equality $I = S$ holds at the goods market equilibrium locus, but the adjustment mechanism is output rather than the profit rate. What these traditions share, and what matters for the accounting laws derived below, is the *constancy* of ϕ^p: no feedback from accumulation dynamics to the investment share operates within the period. The traditions disagree about the causal mechanism that enforces the condition—what adjusts, and through which institutional channel—but they converge on the equilibrium restriction.

The Cambridge equation makes explicit what $\phi^p = s$ represents. Since $\phi^p \equiv I/Y^p$ and $I = s_\pi \Pi$, the investment share decomposes as:

$$\phi^p = s_\pi \cdot \pi^p \tag{8}$$

where s_π is the savings propensity out of profits and $\pi^p \equiv \Pi/Y^p$ is the profit share at capacity. The constant s is therefore the product of a behavioral assumption (the propensity to save out of profits) and a distributional assumption (the share of capacity output appropriated as profit). Both are suppressed—held constant—under the closure imposed here. The constancy of ϕ^p severs the investment share from distributive conflict. This is contradictory: θ governs the choice of technique and its effect on labour productivity, which is itself a site of distributive contestation. The closure holds fixed what the parameter it transmits presupposes as variable within Shaikh (2016)'s own framework.

3.2.4 Accumulation Regimes and Supply-Side Instability

The sign of $(\theta - 1)$ determines whether accumulation is self-reinforcing or self-exhausting. When each unit of capital generates more productive capacity than it displaces ($\theta > 1$), accumulation accelerates its own conditions: each period's investment improves the next period's capital productivity. When each unit generates less capacity than it displaces ($\theta < 1$), accumulation erodes its own conditions: capital productivity falls as the stock grows. The boundary case $\theta = 1$ is the knife-edge where capital and capacity expand in exact proportion and the accumulation rate is stationary. The question is what dynamics each regime produces under the closures derived above.

Differentiating the accounting identity $\hat{k} + \delta = sB$ under constant ϕ^p and substituting the capital productivity growth rate from equation (4) yields a nonlinear ODE of the Bernoulli type[4]:

$$\boxed{\frac{d\hat{k}}{dt} = (\theta(\Lambda) - 1)\,\hat{k}\,(\hat{k} + \delta)} \tag{9}$$

The equation is nonlinear but analytically solvable. The θ-dependent dynamics emerge because the constant investment share prevents the investment channel from absorbing the θ signal—the regime's structural character feeds back directly into the acceleration of accumulation.

The equilibria are $\hat{k}^* = 0$ (net stagnation: gross investment just covers depreciation) and $\hat{k}^* = -\delta$ (zero gross investment: the capital stock is not being replaced). The vector field in the feasible region $\hat{k} > 0$ is governed entirely by the regime, since both $\hat{k}$ and $(\hat{k} + \delta)$ are strictly positive:

Regime	**Condition**	$d\hat{k}/dt$ for $\hat{k} > 0$	**Dynamics**	**Character**
Over-mechanization	$\theta < 1$	negative	$\hat{k}$ decelerates toward 0	Stagnation
Under-mechanization	$\theta > 1$	positive	$\hat{k}$ accelerates toward ∞	Explosion
Balanced growth	$\theta = 1$	zero	$\hat{k}$ constant	Knife-edge

Table 1: Regime classification under the pure unbalanced growth closure.

No feasible interior attractor exists under either regime. Over-mechanization drives capital accumulation toward stagnation—each successive unit of capital generates less capacity, dragging the system toward the point where gross investment merely replaces depreciated stock. Under-mechanization drives capital accumulation toward explosion—each unit generates more capacity, feeding back into further accumulation with no internal brake.

Unbalanced growth is inherently destabilizing on the supply side. The sign of $(\theta - 1)$ determines whether the system decelerates or accelerates, but in neither case does it converge to a feasible

[4]The full derivation is presented in Appendix A.

interior equilibrium. The stabilizing force, when it arrives, must come from the demand side: as $\hat{k}$ rises and the accumulation rate approaches its limits, μ falls below 1, labor markets tighten, the consumption share is squeezed, and the three channels of distribution re-enter simultaneously. The class struggle over distribution, operating through the institutional forms of the accumulation regime, is what prevents the supply-side instability from realizing its tendential law. That terrain is developed in Chapter 2.

The level trajectory of capital productivity confirms the regime classification. From $\hat{b} = (\theta - 1)\hat{k}$ and the solution for $\hat{k}(t)$, integrating gives:

$$B(t) = B_0 \cdot \exp\left[\int_0^t (\theta - 1)\,\hat{k}(\tau)\,d\tau\right] \tag{10}$$

The trajectory separates into a transient component—driven by the initial deviation of $\hat{k}$ from equilibrium, decaying under over-mechanization, exploding under under-mechanization—and a trend component determined by the regime's structural character. Log-differentiating $B(t)$ recovers $\hat{b} = (\theta - 1)\hat{k}$: the mapping is internally consistent.

3.2.5 Mechanization, Autonomous Progress, and the Limits of Shaikh's Closure

The pure closure $\hat{y}^p = \theta\hat{k}$ locates the capital-to-capacity mapping entirely within the mechanization choice governed by $\theta(\Lambda)$—the mechanization possibilities frontier at its origin, where the transformation elasticity is grounded in the connection between surplus extraction and the direction and intensity of technical change. Shaikh's cointegrating relation introduces a second channel:

$$\ln Y = a + bt + d \cdot \ln K \tag{11}$$

which implies, upon differentiation:

$$\hat{y}^p = \theta\,\hat{k} + \beta \tag{12}$$

where β is the rate of autonomous technical progress—capacity growth that is independent of the accumulation rate. The time trend in Shaikh's specification absorbs this term.

The distinction between the pure and extended closures is not merely technical. Under the interpretation developed in this chapter—which Shaikh himself does not articulate, since he does not interpret d as a transformation elasticity of capital into productive capacity—the cointegrating parameter d *is* the empirical counterpart of $\theta(\Lambda)$, and the relation $\hat{y}^p = \theta\hat{k} + \beta$ asserts that unbalanced growth ($\theta \neq 1$) takes place *alongside* a linear deterministic trend representing an autonomous component of technical change. Two features of this specification deserve explicit attention.

Shaikh's empirical specification embeds a claim he does not theorize as such. The transformation of capital into productive capacity operates through two distinct channels simultaneously—a mechanization channel governed by θ and an autonomous channel governed by β. The mechanization channel is regime-dependent by construction; the autonomous channel is, by definition, independent of accumulation, profitability, and the institutional configuration Λ. The coexistence of these two channels within a single cointegrating relation is the empirical structure that the ARDL bounds-testing procedure estimates. It is also the site of an internal contradiction within Shaikh's theoretical framework.

This coexistence sits in tension with Shaikh's own theoretical commitments throughout *Capitalism*. The entire architecture of the book argues for the full endogeneity of technical change to profitability: the profit rate drives the direction and intensity of mechanization, investment responds to profitability signals, and the classical competitive process selects techniques on the basis of cost-cutting driven by the wage-profit frontier (Shaikh, 2016). There is no theoretical

location in that architecture for a component of capacity growth that operates independently of the accumulation rate and the profit-driven mechanization channel. Yet the time trend in his cointegrating relation is precisely such a component. The contradiction is invisible to Shaikh because he does not interpret d as unbalanced growth—the parameter remains an unexplained long-run elasticity rather than a structural object whose coexistence with β requires theoretical justification.

This chapter does not resolve the contradiction. It inherits it: the critical replication operates within Shaikh's extended closure because the empirical strategy requires the time trend to absorb secular technical progress. Whether the autonomous channel can be grounded theoretically—or whether it is an artifact of a closure that suppresses a distributional channel the theory requires—is a question this chapter's empirical findings will sharpen but cannot settle. Here, the consequences of Shaikh's extended closure are derived in order to locate the quadratic structure that his specification implies.

Under $\phi^p = s$ and $\hat{y}^p = \theta\hat{k} + \beta$, the capital productivity growth rate becomes $\hat{b} = (\theta - 1)\hat{k} + \beta$, and the ODE is:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta)\left[(\theta - 1)\hat{k} + \beta\right] \tag{13}$$

Expanding:

$$\frac{d\hat{k}}{dt} = (\theta - 1)\hat{k}^2 + [(\theta - 1)\delta + \beta]\,\hat{k} + \beta\delta \tag{14}$$

This is a genuine quadratic in $\hat{k}$, with equilibria at the roots of $(\hat{k} + \delta)[(\theta - 1)\hat{k} + \beta] = 0$:

$$\hat{k}_1^* = -\delta, \qquad \hat{k}_2^* = \frac{\beta}{1 - \theta} \tag{15}$$

The second equilibrium is new: it is the net growth rate at which autonomous technical progress exactly offsets the regime's structural drag. Under over-mechanization ($\theta < 1$) and $\beta > 0$: $\hat{k}_2^* = \beta/(1 - \theta) > 0$—a feasible positive equilibrium emerges. Autonomous technical progress provides the interior attractor that the pure closure could not generate.

Writing the quadratic $(\theta - 1)\hat{k}^2 + [(\theta - 1)\delta + \beta]\hat{k} + \beta\delta = 0$, Vieta's formulas give:

$$\hat{k}_1 \cdot \hat{k}_2 = \frac{\beta\delta}{\theta - 1} \tag{16}$$

$$\hat{k}_1 + \hat{k}_2 = -\delta - \frac{\beta}{\theta - 1} \tag{17}$$

Under over-mechanization ($\theta < 1$) the product of roots is negative—one positive root, one negative—confirming the existence of a feasible attractor alongside a decumulation equilibrium. Under under-mechanization ($\theta > 1$) the product is positive and both roots share sign. The depreciation rate anchors the sum; the ratio $\beta/(\theta - 1)$ scales autonomous progress by the inverse degree of unbalanced growth.

As $\theta \to 1$, the quadratic coefficient vanishes and the equilibrium $\hat{k}_2^*$ diverges:

$$\mathcal{D} \equiv \{\theta(\Lambda) = 1, \;\; \beta = 0\} \tag{18}$$

At $\mathcal{D}$ both the regime classification and the autonomous-progress channel collapse simultaneously—the system has no determinate growth rate. The degeneracy locus is a singularity in the (θ, β) parameter plane.

When $\beta \to 0$, $\hat{k}_2^* \to 0$, the positive equilibrium collapses to the stagnation point, the quadratic degenerates to $(\theta - 1)\hat{k}(\hat{k} + \delta) = 0$, and the Bernoulli ODE of §3.2.4 holds. The pure closure is the limiting case of Shaikh's specification in which the entire capital-to-capacity mapping is exhausted by the mechanization choice. The quadratic structure and Vieta's formulas are properties of Shaikh's extended closure, not of the accounting laws per se.

3.2.6 Summary: What the Accounting Laws Establish

The regime classification, the instability result, the capital productivity mapping, and the absence of a feasible interior attractor all emerge from three ingredients: the accounting identity $\hat{k}+\delta = \phi^p B$, the unbalanced growth closure $\hat{y}^p = \theta\hat{k}$, and the investment-savings closure $\phi^p = s$. The first two are pre-behavioral; the third is the only behavioral assumption in the system. No investment function, Phillips curve, or demand-formation equation has been imposed.

The **regime classification** is determined by the sign of $(\theta - 1)$. Over-mechanization ($\theta < 1$) generates decelerating accumulation; under-mechanization ($\theta > 1$) generates accelerating accumulation; balanced growth ($\theta = 1$) is the knife-edge singularity.

No feasible interior attractor exists under the pure closure. The supply-side accounting structure is inherently destabilizing. The demand side provides the stabilizing force.

The **quadratic structure**—Vieta's formulas, the degeneracy locus, and the feasible interior equilibrium under over-mechanization—requires Shaikh's extended closure $\hat{y}^p = \theta\hat{k} + \beta$. The autonomous progress term β introduces a channel absent from the mechanization possibilities frontier at its origin. This is a finding, not merely a technicality: it means the quadratic law is a property of Shaikh's empirical specification, not of the pure accounting laws, and locates it within his identification strategy. The coexistence of $\theta \neq 1$ and $\beta > 0$ in a single cointegrating relation embeds a theoretical tension between the regime-dependent mechanization channel and an autonomous component of technical change that has no grounding within Shaikh's own commitment to the full endogeneity of technical change to profitability.

The empirical task is now defined. The transformation elasticity $\theta(\Lambda)$ must be recovered from observed data, and its regime-dependence tested. The next section develops Shaikh's identification strategy for this recovery and names its limitation; the section that follows subjects that strategy to critical replication.

3.3 Shaikh's Triple Contribution: Instability, Correction, Portability

The accounting laws derived in the previous section establish that unbalanced growth is inherently destabilizing on the supply side: under over-mechanization, accumulation decelerates toward stagnation; under under-mechanization, it accelerates without bound. No feasible interior attractor exists under the pure closure. This is Shaikh's first contribution to the problem of capacity utilization, although he does not frame it in these terms. The instability result inverts the standard macroeconomic intuition: it is the supply side that destabilizes, and the demand side—operating through utilization, labor markets, and the distributional conflict—that provides the stabilizing force. The institutional forms through which that stabilization operates are regime-specific, which is why the transformation elasticity θ cannot be regime-invariant if the theory is taken seriously. The instability result is the theoretical foundation for the empirical recovery of $\theta(\Lambda)$: the parameter matters because it governs the structural character of accumulation dynamics, not merely the statistical fit of an output–capital regression.

Shaikh's second contribution is the construction of an empirical identification strategy that translates the instability result into an estimable cointegrating relation. The identification chain proceeds in four steps. First, the aggregate profit rate is decomposed into a product of the profit share, capacity utilization, and capital productivity: $r = \pi \cdot u \cdot B$. Second, a technical progress function links capital productivity growth to the rate of accumulation through a mechanization channel and an autonomous component, yielding the growth-rate relation $\hat{y}^p = \theta\hat{k} + \beta$ derived in §3.2.5. Third, the classical long-run closure $\mu = 1$ pins the theoretical anchor: the long-run component of the output–capital relation is interpreted as the trajectory of productive capacity, with utilization recovered as the residual deviation from that trajectory. Fourth, the growth-rate

relation is integrated to yield the log-level cointegrating relation:

$$\ln Y_t = a + bt + d \cdot \ln K_t + \varepsilon_t \tag{19}$$

where d—the empirical counterpart of $\theta(\Lambda)$—is the long-run output–capital elasticity, b absorbs the autonomous progress term β, and ε_t captures the non-cointegrated component—capacity utilization under the maintained closure. Shaikh operationalizes this relation using the ARDL bounds-testing framework of Pesaran et al. (2001), estimating it in error-correction form:

$$\Delta y_t = \alpha + \sum_{i=1}^{p-1} \psi_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \varphi_j \Delta k_{t-j} + \phi_1 y_{t-1} + \phi_2 k_{t-1} + \gamma t + \sum_h \delta_h DM_{h,t} + \varepsilon_t \tag{20}$$

where y_t and k_t are log output and log capital deflated by a common price index—a requirement Shaikh derives from the profit-rate-consistent identification constraint—and $DM_{h,t}$ is a structure of historical dummies. Conditional on the bounds F-test rejecting the null of no long-run relation, the long-run multipliers are recovered from the OLS estimates, and the capacity benchmark is constructed as $\widehat{y_t^p} = \hat{a} + \hat{d} \cdot k_t + \sum_h \hat{c}_h DM_{h,t}$. Shaikh's reported benchmark values are $\hat{d} = 0.6609$ and a sample length of $T = 61$ annual observations for the US corporate sector (1947–2007).

Shaikh's third contribution is portability. The identification strategy requires only national accounts data—gross value added, fixed capital stock, and a price deflator—all of which are available in any economy with a minimally functioning statistical office. It does not require industrial surveys of the kind administered by the Federal Reserve Board, which are confined to the US manufacturing sector and have no international counterpart of comparable scope. It does not require output-gap filters, which impose stationarity assumptions that erase the structural signal the cointegrating relation is designed to recover. The portability of the method is what makes the Chilean application possible as a structurally distinct test of the same identification strategy under peripheral conditions: commodity dependence, financial subordination, and structural-balance fiscal governance. It is also what gives the filter comparison developed later structural content rather than reducing it to a horse race between alternative smoothing devices.

The limitation of the identification strategy is equally specific. The ARDL framework recovers d as a fixed parameter—a single scalar summarizing the long-run output–capital relation across the entire sample. The estimation cannot register the regime-dependence of $\theta(\Lambda)$ that the accounting laws predict. The theory developed in §3.2 says that the transformation elasticity is a property of the institutional configuration: it should shift when the accumulation regime shifts. The empirical strategy treats it as constant. This is not a deficiency of the ARDL procedure as such—bounds testing is agnostic about parameter stability—but of the specific implementation, which imposes constancy without testing it. The gap between what the theory predicts (regime-dependence) and what the estimation assumes (constancy) is the specific site where critical replication intervenes. The question is not whether Shaikh's cointegrating relation holds, but whether the fixed parameter it recovers is a structural feature of the data or a consequence of imposing constancy on a parameter that varies across the accumulation regimes the method was designed to characterize.

3.4 Closure, Identification, and the Case for Critical Replication

? diagnoses Shaikh's *Capitalism* as relying on closed-system abstraction that suppresses historical contingency—an ontological critique that is correct at its level but incomplete without an operative method. The problem is not modelling: the accounting laws derived in this chapter are formal models, and they work. The problem is *closure*—which variables are held fixed by

assumption, and whether what is suppressed is essential to the object under investigation. Nikiforos (2020) provides the sharper instrument: closure is a special form of abstraction, evaluable by two criteria—it must posit mechanisms that are *real* (subject to empirical scrutiny) and isolate what is *essential* (the primary determinants, not merely the general). A closure that holds constant what the theory itself identifies as a primary determinant does not fail because it is formal, but because it formalizes in the wrong place. Nor is the problem mathematics: it is *identification*—whether the econometric procedure recovers the structural object the theory names or a weaker reduced-form relation that absorbs the structural variation into a fixed parameter.

These distinctions become operative through empirical replication as a researcher practice. Herndon et al. (2014) demonstrated that Reinhart and Rogoff's canonical finding—a debt-to-GDP threshold above which growth collapses—depended on a fixed summary statistic that masked compositional heterogeneity across countries and time periods. The stakes were not academic: the finding had been escalated to supranational institutions and directly informed the austerity prescriptions imposed on debtor economies throughout the post-crisis period, lending scientific authority to a causal claim between public debt and growth that the underlying data did not support. What Herndon, Ash, and Pollin showed was that the object itself—the threshold, the causal direction, the universality of the result—dissolved under reconstruction. The scalar $d = 0.6609$ in Shaikh's identification faces a structurally analogous risk. If the transformation elasticity varies across accumulation regimes—Fordist versus post-Fordist, core versus periphery—then a fixed-parameter estimate averages over that variation, producing a single number whose economic interpretation is ambiguous: it could represent the structural character of any one regime, or none. Empirical replication that merely verifies the computation leaves this risk untouched. Critical replication reconstructs the empirical object itself—subjecting its closure, its identification, and its measurement to the standard that critical realism articulates but cannot, by ontological diagnosis alone, enforce. It is replication as a weapon of contestation and plurality in economic thinking: critical realism operationalized through econometrics, not beyond mathematics but through better identification.

Shaikh's empirical strategy imposes two closures. The first—the classical long-run closure $\mu = 1$, in which demand meets supply and all output is realized—is an appropriate abstraction: it suppresses the demand side in order to isolate the supply-side accounting structure, and the accounting laws of §3.2 demonstrate that the suppressed material can be recovered when the abstraction is lifted. Combined with unbalanced growth ($\theta \neq 1$), this closure produces smooth reproduction—the benchmark from which the accounting laws are derived and from which μ can be reintroduced. The second closure is the treatment of θ as a fixed parameter across the full sample. This is the closure that fails the essentiality criterion. Shaikh's own theoretical architecture in *Capitalism*—endogenous technical change driven by profitability, Harrodian instability, the regime-dependence of accumulation dynamics—identifies the transformation elasticity as a property of the institutional configuration, not a timeless constant. The accounting laws derived in this chapter formalize that dependence as $\theta(\Lambda)$. An estimation strategy that holds θ fixed suppresses what the theory itself designates as essential: the variation of the capital-to-capacity mapping across accumulation regimes. The problem is not that Shaikh formalizes—the entire conceptual framework of this chapter is formalization—but that his specific closure choice absorbs the regime signal into a constant, making it empirically invisible.

4 Critical Replication Studies on Shaikh's Approach to Cointegration

4.1 Confinement under Uncertainty

The critical replication confines Shaikh's identification along two axes: specification uncertainty and estimator structure. The question is whether the capacity anchor is a property of the data or of the researcher's choices. Each stage opens where the previous one broke.

4.2 The Measurement Apparatus: Shaikh's Generalized Perpetual Inventory Method

The capital stock series entering the identification are not raw BEA aggregates. They are constructed through Shaikh's Generalized Perpetual Inventory Method (GPIM), which closes a gap in the national accounts: the standard micro-level perpetual inventory identity does not survive aggregation under the Fisher-ideal chain-weighting that the BEA uses to construct aggregate capital stocks.[5]

The GPIM accumulation rule for the aggregate current-cost capital stock is:

$$K_t \;=\; IG_t \;+\; z_t^* \cdot K_{t-1}, \qquad z_t^* \;\equiv\; (1 - z_t)\,\frac{p_t^K}{p_{t-1}^K}, \tag{21}$$

where IG_t is aggregate gross investment in current prices, z_t is the aggregate depletion rate, and z_t^* compresses two conceptually distinct operations into a single coefficient: the fraction $(1 - z_t)$ of the stock that physically survives the period, and the revaluation of that surviving stock from period-$(t-1)$ to period-t prices. Survival is physical and comes first; revaluation is the accounting price adjustment applied to whatever survived. The two operations commute algebraically, but conceptually the distinction matters: the gross stock counts all surviving machines equally (depletion governs exit), while the net stock discounts them by accumulated value attrition (depreciation governs book value). Productive capacity depends on the physical quantity of installed capital, not on its accounting valuation — a machine 80% depreciated still produces at full service capacity until it retires. The gross stock is therefore the theoretically appropriate capacity measure; the net stock confounds a price concept with a physical concept.

For the US corporate sector, Shaikh applies three adjustments to the BEA official series: substitution of 1993-vintage BEA depletion rates for the post-revision geometric rates, pre-war initial values from IRS balance-sheet data rather than BEA backward extrapolation, and a Great Depression write-down correction that reduces the 1947 corporate net stock by approximately 28% relative to the BEA 2011 official figure. The resulting series — KGCcorp_t (gross, GPIM-adjusted) and KNCcorp_t (net, GPIM-adjusted) — together with the BEA benchmark KNCcorpbea_t, enter the replication as the capital-stock objects. Full construction formulas and the GPIM derivation from the micro-level identity through the aggregation problem to the corporate adjustments are provided in Appendix **??**.

[5] The data objects are drawn from Appendix 6.8 of Shaikh (2016), accessible at the companion website `realeconomicanalysis.com`. The source spreadsheet is `_Appendix6.8DataTablesCorrected.xlsx`. A data audit verifying series alignment is documented in script `90_data_audit.R` (output in `output/CriticalReplication/data_audit/`).

4.3 Data

Shaikh does not document which NIPA series correspond to the regression variables labeled `l_ys` and `l_ks` in Appendix Table 6.7.14. No replication dataset accompanies the published work. The replicator must reverse-engineer the data construction from indirect evidence—coefficient magnitudes, sample size, fit statistics—rather than from an explicit data manifest.

The replication uses net value added of the corporate sector (VAcorp_t) rather than gross value added (GVAcorp_t). This correction is not an arbitrary switch chosen for better results; it is grounded in the GPIM formalization of the four output–capital pairings and can be stated as a theorem about residual structure. The capacity closure defines productive potential with respect to the gross capital stock:

$$Y_t^p = A_0\, e^{bt}\, \left(\tilde{K}_t^{G,R}\right)^{\theta(\Lambda)} \tag{22}$$

where $\tilde{K}_t^{G,R} = K_t^G/p_t$ is the gross stock under the common GDP deflator. All four logically possible output–capital pairings recover the same elasticity θ under this closure, but they differ in residual composition. The output identity $\text{NVA}_t = \text{GVA}_t(1 - s_t^D)$ and the stock identity $\ln \tilde{K}_t^N = \ln \tilde{K}_t^G + \ln \psi_t$ generate each pairing's error term by substitution:

$$\varepsilon_t^{GG} = \ln u_t, \qquad \varepsilon_t^{NG} = \ln u_t + \ln(1-s_t^D), \qquad \varepsilon_t^{NN} = \ln u_t + \ln(1-s_t^D) - \theta \ln \psi_t, \qquad \varepsilon_t^{GN} = \ln u_t - \theta \ln \psi_t, \tag{23}$$

where $s_t^D \equiv D_t/\text{GVA}_t$ is the depreciation share and $\psi_t \equiv K_t^N/K_t^G$ is the net-to-gross ratio. Cointegration requires each residual to be $I(0)$. Since s_t^D trends upward (secular rise in CFC/GVA under post-2013 BEA data releases) and ψ_t trends downward (shift toward shorter-lived assets), the pairings that load these trending variables into the residual require progressively stronger auxiliary assumptions: GG requires none beyond the closure; NG requires s_t^D trend-stationary; NN requires both s_t^D and ψ_t trend-stationary; GN requires ψ_t trend-stationary. The full derivation with proofs is in Appendix **??**.

$\text{NVA} \sim K^G$ is therefore the only pairing satisfying three criteria simultaneously: (i) the correct capacity concept on the right-hand side (gross stock = physical installed capacity), (ii) stock-flow consistent output on the left-hand side (NVA nets out the depreciation flow mechanically tied to gross stock), and (iii) no dependence on the net-to-gross ratio's stationarity. The four-pairing exercise reported in §4.4.1 indicates empirically that the capital concept dominates the specification sensitivity (25–85% θ swing governed by K^G versus K^N), while the output measure contributes less than three percentage points within any capital definition.

Figure 1: Depreciation share $s_t^D = D_t/\text{GVA}_t$ (left) and net-to-gross ratio $\psi_t = K_t^{N,R}/K_t^{G,R}$ (right), US corporate sector, 1947–2011. Linear trend overlays. The depreciation share deviations are small and stationary (flow ratio); the net-to-gross deviations are slow and persistent (stock ratio).

The transformed series entering all replication stages are $y_t \equiv \ln(\text{VAcorp}_t/p_t)$ and $k_t \equiv \ln(\text{KGCcorp}_t/p_t)$, where p_t is the GDP implicit price deflator (BEA NIPA Table 1.1.9, line 1). The common-deflator rule is a theoretical commitment: it eliminates the relative price channel between output goods and capital goods by construction, preserving the profit-rate-consistent identification derived in Shaikh (2016, Appendix 6.6). One tension internal to this specification must be noted. The common-deflator real stock $\tilde{K}_t = K_t/p_t$ does not obey the GPIM real accumulation rule, which governs $K_t^R = K_t/p_t^K$. The two objects differ by the relative price of capital goods: $\tilde{K}_t = (p_t^K/p_t) \cdot K_t^R$. The common deflator preserves ratio consistency at the cost of breaking the stock's own law of motion — a confinement limit of Shaikh's data environment, not an invalidation of the identification. Full construction formulas, source-line provenance, and release notes are in Appendix B.

Sample: 1947–2011 ($T = 65$), with effective estimation sample determined by the lag order. The three step dummies d_{1956}, d_{1974}, d_{1980} follow Shaikh's periodization. In S0, these enter the long-run equation as permanent level shifts; in S1, they are treated as transitory (impulse dummies, short-run only) or omitted, consistent with PSS case discipline.

The canonical initial-year gross capital stock is $\mathrm{KGC}^{corp}_{1947} = 170.58$ Bn, reconstructed via the Generalized Perpetual Inventory Method. Shaikh does not report this value; discrepancies between the replication's initial value and Shaikh's propagate through the entire capital stock series and into the ARDL coefficients.

A fourth variable, the corrected corporate profit share $\mathrm{Profshcorp}_t$, is constructed in the same data environment as the empirical proxy for the rate of exploitation $\ln e_t$ entering the $m = 3$ VECM system in Stage S2. It does not enter the S0 or S1 identification objects.

One replication gap must be noted: Shaikh's Appendix 6.8 tables do not tabulate the output-side deflator used in the corporate utilization exercise, leaving p_t implicit.[6] The GDP implicit price deflator (BEA NIPA Table 1.1.9, line 1) is used as the common numeraire—the minimal transparent choice consistent with the identification constraint. Non-zero RMSE between the replicated and published utilization series may therefore reflect this deflator gap rather than a specification error.

The estimated $\hat{\theta}$ recovered from the ARDL long-run multiplier is an operational proxy for the transformation elasticity $\theta(\Lambda)$, not a direct measure of it. The proxy relationship depends on the validity of the capacity closure (22), the common-deflator rule, and the stationarity of the cointegrating residual.

4.4 Stage S0: Faithful Replication of the Capacity Benchmark

The replication saturates Shaikh's identification device rather than replacing it, proceeding in three stages. Stage S0 reconstructs the capacity benchmark as faithfully as possible under a fixed specification: the data objects, the common deflator, and the ARDL representation are all held at Shaikh's reported values, and the benchmark is recovered without contesting any design choice. Stage S1 embeds this fixed point in an admissible ARDL specification lattice and asks whether the recovered benchmark is stable across the neighbourhood of that point. Stage S2 re-estimates the same long-run restriction as a system property using a VECM, closing the sufficiency gap that single-equation stability leaves open. The pipeline is: corporate accounting objects → common deflator transformation → ARDL estimation under the fixed specification → bounds admissibility screening → long-run coefficient recovery → productive capacity benchmark → utilization residual.

Shaikh's identification rests on two elements. The first is a maintained long-run closure in which utilization converges to its normal level ($u = 1$), fixing the theoretical anchor of the output–capital relation: under this closure, the long-run component of the output–capital relation is interpreted as the trajectory of productive capacity. The second is the set of long-run multipliers recovered from the ARDL error-correction representation, which operationalize that anchor empirically by translating the closure into an estimable cointegrating relation. Stage S0 holds both elements fixed. What S0 establishes is the baseline empirical object—the productive capacity benchmark $\widehat{\ln Y^p_t}$ and the implied utilization series $\widehat{u}_t$—under Shaikh's exact specification, together with the admissibility status of that identification under his own methodology.

Shaikh operationalizes the capacity benchmark by estimating the output–capital relation in

[6] The Appendix 6.8 spreadsheet includes an internal note cross-referencing Appendix 6.7.VI but states that the matching utilization-construction data are not tabulated.

ARDL error-correction form:

$$\Delta y_t = \alpha + \sum_{i=1}^{p-1} \psi_i \, \Delta y_{t-i} + \sum_{j=0}^{q-1} \varphi_j \, \Delta k_{t-j} + \phi_1 y_{t-1} + \phi_2 k_{t-1} + \gamma t + \sum_h \delta_h DM_{h,t} + \varepsilon_t, \tag{24}$$

estimated by OLS via the `ARDL` package for R (Natsiopoulos and Tzeremes, 2024). The benchmark specification is fixed at $m_0 = (p_0, q_0, c_0, s_0) = (2, 4, \text{III}, \{d56, d74, d80\})$: lag orders ARDL(2,4), deterministic Case III (unrestricted intercept, no trend in the long-run relation), and Shaikh's reported step-dummy structure. This specification is a single point in the admissible ARDL space that Stage S1 will map systematically; it is flagged as the Shaikh benchmark point throughout. Admissibility is assessed at three levels. At the primary level, the bounds F-test examines whether the lagged level terms jointly identify a long-run relation:

$$H_0 : \ \phi_1 = \phi_2 = 0. \tag{25}$$

If H_0 is rejected, a non-degenerate long-run relation is identified and the capacity benchmark is admissible. If H_0 is not rejected across all five PSS cases, S0 acknowledges operating as a filter rather than as a cointegrating estimator—the utilization series is still constructed but its inferential status is degraded. At the secondary level, the bounds t-test on the lagged error-correction term provides a stricter robustness diagnostic where case-specific critical values are defined (Cases III and V); it is not an admissibility gate. At the replication level, RMSE between the recovered $\widehat{u}_t$ and Shaikh's published series (Appendix Figure 6.6.1) assesses whether the S0 object faithfully reproduces the canonical benchmark.

The bounds framework distinguishes five deterministic cases according to whether intercept and trend are excluded, restricted to the long-run relation, or left unrestricted in the short-run dynamics. Table 2 reports the implied long-run benchmark for each case. Cases II and III, and likewise Cases IV and V, imply the same compact benchmark form; their difference lies in whether the deterministic components are restricted to the long-run relation or left unrestricted in the short-run ARDL representation. Shaikh's benchmark specification corresponds to Case III ($\star$).

Table 2: Deterministic cases in the ARDL bounds framework

Case	Deterministic treatment	Long-run benchmark $\widehat{y_t^p}$	Admissibility gate	Robustness diagnostic
I	No intercept, no trend	$\widehat{\theta}\, k_t$	F-bounds	—
II	Restricted intercept, no trend	$\widehat{\mu} + \widehat{\theta}\, k_t$	F-bounds	not defined
III$\star$	Unrestricted intercept, no trend	$\widehat{\mu} + \widehat{\theta}\, k_t$	F-bounds	t-bounds
IV	Unrestricted intercept, restricted trend	$\widehat{\mu} + \widehat{\theta}\, k_t + \widehat{\delta}\, t$	F-bounds	not defined
V	Unrestricted intercept, unrestricted trend	$\widehat{\mu} + \widehat{\theta}\, k_t + \widehat{\delta}\, t$	F-bounds	t-bounds

$\star$ Shaikh's benchmark case: ARDL(2,4), Case III, step dummies $\{d56, d74, d80\}$.

The data objects entering the S0 identification are y_t, k_t, and the fixed step-dummy structure $\{d56, d74, d80\}$. Construction of GVAcorp_t, KGCcorp_t, and p_t is described in §4.3. A distribution proxy, Profshcorp_t, is constructed in the same data environment for later system-stage use but does not enter the S0 identification object.

Conditional on admissibility, the long-run multipliers are recovered from the OLS estimates via

the standard ARDL reparameterization:

$$\hat{a} = \frac{\hat{\alpha}}{1 - \sum_i \hat{\psi}_i}, \qquad \hat{\theta} = \frac{\sum_j \hat{\varphi}_j}{1 - \sum_i \hat{\psi}_i}, \qquad \hat{c}_h = \frac{\hat{\delta}_h}{1 - \sum_i \hat{\psi}_i}. \tag{26}$$

The productive capacity benchmark is then:

$$\widehat{y_t^p} = \hat{a} + \hat{\theta}\, k_t + \sum_h \hat{c}_h\, DM_{h,t}, \tag{27}$$

and utilization is recovered residually as:

$$\widehat{u}_t \;\equiv\; \exp\Big(y_t - \widehat{y_t^p}\Big)\,. \tag{28}$$

The Shaikh benchmark values from Table 6.7.14 are: $\hat{\theta} = 0.6609$, $\hat{a} = 2.1782$, and long-run dummy multipliers $\hat{c}_{d74} = -0.8548$, $\hat{c}_{d56} = -0.7428$, $\hat{c}_{d80} = -0.4780$. The short-run capital memory share at the benchmark specification is $s_K = q_0/(p_0 + q_0) = 4/6 = 0.667$, meaning that two-thirds of the short-run dynamic budget is allocated to capital lags. The replication quality is assessed by computing RMSE between $\widehat{u}_t$ and Shaikh's published corporate utilization series; non-zero RMSE at this stage may reflect the deflator gap rather than a specification error.

Stage S0 establishes a single empirical object. The transition to Stage S1 is structured around four moves. First, the five-case deterministic exploration: S0's Case III selection is one of five admissible cases in the PSS framework, each implying a different long-run benchmark; before expanding to a full specification lattice, the admissibility status and RMSE across all five cases are reported, checking whether the Case III selection is robust to the deterministic treatment. Second, replication visualization: the comparison between $\widehat{u}_t$ and Shaikh's published series licenses the claim that the S0 object faithfully reproduces the canonical benchmark before any specification exploration begins. Third, RMSE ranking across the five cases provides a data-driven proximity metric—not a model-selection criterion—that assesses whether the deterministic choice is innocuous. Fourth, and most importantly, the AIC and BIC coordinates of the benchmark specification locate it as a single point in the fit–complexity plane. Stage S1 asks whether the capacity anchor is a property of a *family* of specifications occupying the neighbourhood of that point, rather than a property of the particular point that Shaikh's lag-selection procedure selected.

4.4.1 S0 Results: Non-Replication

No coefficient, fit statistic, or long-run multiplier from Shaikh's Appendix Table 6.7.14 replicates under current-release BEA data. The discrepancy is not marginal. Shaikh's capital stock coefficients (L0 through L4) are an order of magnitude larger than the replication's, and all are individually significant at the 1% level; the replication finds significance only at L0 and L1. Shaikh's RMSE (0.016) is roughly half the replication's (0.031). The long-run elasticity θ is in the right neighbourhood—0.661 (Shaikh) versus 0.745 (replication, no trend)—but the path to it is entirely different: different autoregressive structure, different denominator, different dummy effects.

The most likely sources are data release differences (the BEA's comprehensive revisions, especially the 2013 R&D capitalization), Shaikh's undocumented variable definitions and deflator choice, and the unreported initial capital stock value. These are reproducibility failures, not specification failures—they concern the inputs to the method, not the method itself. The non-replication concerns point estimates, not the existence of the cointegrating relationship. Whether output and capital maintain a long-run equilibrium under current-release data is a separate question, tested in the four-pairing exercise and the PSS bounds test below.

Table 3: ARDL(2,4) coefficient comparison: Shaikh (2016) versus replication

Variable	Shaikh (2016)	Replication (no trend)	Replication (with trend)
Lagged dependent ($\ln Y$)			
L1	1.298^{***}	0.891^{***}	0.884^{***}
L2	-0.393^{***}	-0.015	-0.016
Capital stock ($\ln K$)			
L0	5.086^{***}	0.753^{**}	0.755^{**}
L1	-14.285^{***}	-1.143^{***}	-1.157^{***}
L2	15.705^{***}	0.057	0.063
L3	-8.296^{***}	0.286	0.283
L4	1.853^{**}	0.139	0.128
Dummies			
d_{74}	-0.081^{***}	-0.013	-0.008
d_{56}	-0.071^{***}	-0.012	-0.007
d_{80}	-0.045^{**}	-0.010	-0.009
Deterministic			
Trend	—	—	0.001
Constant	0.207^{**}	0.257^{*}	0.462

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Sample: 1951–2011 ($N = 61$). Replication uses NVA $\sim K^G$ with current-release BEA data.

4.4.2 S0 Results: The Four-Pairing Exercise

To determine whether the specification sensitivity documented at the NVA $\sim K^G$ baseline is an artifact of the particular pairing, the ARDL(2,4) is re-estimated across all four output–capital combinations, each with and without an explicit linear trend. Eight specifications total, all other settings identical.

Table 4: Four-pairing robustness: θ, π_y, and $1 - \Sigma\gamma$ across output–capital definitions

Pairing	Spec	θ	π_y	$1 - \Sigma\gamma$
GVA $\sim K^G$	no trend	0.750	-0.089^{*}	0.111
GVA $\sim K^G$	trend	0.565	$+0.013$	0.119
NVA $\sim K^G$	no trend	0.745	-0.096^{*}	0.124
NVA $\sim K^G$	trend	0.541	$+0.013$	0.132
NVA $\sim K^N$	no trend	0.807	-0.052^{*}	0.063
NVA $\sim K^N$	trend	0.129	-0.029	0.089
GVA $\sim K^N$	no trend	0.769	-0.046^{*}	0.053
GVA $\sim K^N$	trend	0.111	-0.046	0.077

$^{*}p < 0.05$ on the RECM error-correction coefficient. All three dummies individually insignificant ($p > 0.5$) across all eight specifications.

The table partitions into two blocks governed by the capital stock concept, not the output measure. The gross K block ($1 - \Sigma\gamma \approx 0.11$–$0.13$) shows θ dropping approximately 25% upon trend inclusion, with π_y flipping from negative-significant to positive-insignificant. The net K block ($1 - \Sigma\gamma \approx 0.05$–$0.09$) shows θ collapsing approximately 85%, with π_y retaining its sign but losing all statistical significance. Interpreted through the measurement framework, these results suggest that the ARDL long-run multiplier is not identified as a stable structural parameter at this estimation site—it is a specification-contingent projection through a near-zero denominator, whose severity is predictable from the capital stock concept used.

Figure 2: Implicit capacity utilization: 8 pairings versus benchmarks, US corporate sector, 1947–2011. Left: No trend. Right: With trend. Blue: Gross K. Orange: Net K. Dark red: Shaikh u_K. Black: FRB.

4.4.3 S0 Results: Persistence Amplification

The instability has a precise mechanism. The ARDL long-run multiplier $\theta = \Sigma\hat{\phi}_i/(1 - \Sigma\hat{\gamma}_i)$ is a ratio in which the denominator measures how fast output forgets its own past—the memory decay rate of the autoregressive process. When the denominator is small, output has long memory: shocks persist, the "long run" to which the multiplier extrapolates is distant, and the extrapolation is sensitive to anything that perturbs the allocation of trending variation between the autoregressive lags and the capital stock coefficients.

A trend term competes with $\ln K$ for the same upward-trending variation in a postwar sample where capital grows monotonically. The insignificant trend carries no information; it merely reshuffles weight between collinear regressors. A structural parameter should be invariant to such a reshuffling. θ is not, because the persistence of output amplifies the specification perturbation into a qualitative change in the long-run projection. The persistence amplification documented here is the econometric counterpart of the long-memory amplification that the accounting laws predict when $\theta < 1$: in the over-mechanization regime, the output–capital system has a weak attractor and long memory, and the ARDL's denominator sits close to zero as a direct consequence.

A note on interpretation is warranted. This persistence amplification is a property of the ARDL estimator at this site. Whether it reflects genuine long-memory dynamics in the output–capital relationship or an artifact of near-unit-root estimation cannot be resolved within the ARDL framework alone. The VECM estimator in S2 recovers θ as an eigenvector of a reduced-rank regression—a fundamentally different approach that does not pass through the near-zero divisor.

4.4.4 What S0 Establishes

S0 documents a fragility architecture that persists across every specification and variable pairing tested. The method's core pathologies—the persistence amplification of specification choices through the near-zero denominator, the trend sensitivity, the capital-concept regime—are structural properties of the ARDL long-run multiplier at this estimation site, not artifacts of Shaikh's particular data release or variable definitions.

Three questions remain open: (O1) is the specification arbitrary? (O2) which PSS case governs the bounds test? (O3) does the normalization assumption bias the utilization series? S1 addresses O1 and O2 by releasing the specification axis.

4.5 Stage S1: ARDL Specification Geometry

Stage S0 established the output–capital relation as the empirical site of Shaikh's capacity identification, recovering a long-run benchmark under a fixed specification tuple (p_0, q_0, c_0, s_0): lag orders on output and capital, a deterministic case, and a step-dummy structure. What S0 did not establish is whether the long-run multipliers and the implied utilization series are properties of the data or of that particular specification choice. The capacity benchmark was recovered from one point in the admissible ARDL specification space. S1 asks whether the benchmark survives when the specification is allowed to vary across that space.

Shaikh's identification rests on two elements. The first is a maintained long-run closure in which utilization converges to its normal level ($u = 1$), fixing the theoretical anchor of the output–capital relation. The second is the set of long-run multipliers recovered from the ARDL error-correction representation, which operationalize that anchor empirically by translating the closure into an estimable cointegrating relation. S1 holds the first element fixed: the closure $u = 1$ is not under interrogation. What S1 interrogates is the second element—whether the long-run multipliers, and the utilization series they imply, are stable across the admissible specification space or contingent on a single specification selection.

The question is framed as follows. Standard practice selects a single ARDL specification by minimizing an information criterion—AIC, BIC, or a variant—and treats the resulting model as the basis for inference. This procedure embeds a researcher prior: the choice of criterion, and the implicit weighting of fit against parsimony that each criterion encodes, conditions the selected specification. S1 does not ask which specification is the true model. It asks whether the capacity anchor is recoverable as a property of a *family* of specifications occupying the global Pareto frontier of fit and parametric complexity, rather than as a property of any single winner. If the anchor is stable across that family, the identification claim does not depend on the researcher's choice of information criterion. If the anchor is unstable, the identification is conditional on a prior that has no independent justification.

The specification space is organized as a lattice. Each candidate model $m \in \mathcal{L}_{S1}$ is indexed by a four-dimensional tuple:

$$\mathcal{L}_{S1} \;=\; \{\, (p,\, q,\, c,\, s) \,\}, \tag{29}$$

where p is the lag order on output, governing the autoregressive dynamics attributed to the dependent variable; q is the lag order on capital, governing how many lagged levels and differences of the forcing variable enter the conditional model; c indexes the deterministic case (Cases I–V in the classification of Pesaran et al. 2001), which determines whether an intercept and trend appear in the long-run relation, the short-run dynamics, or both; and s indexes the step-dummy structure, capturing historical episodes whose placement is a maintained specification choice.[7]

Not every element of the lattice is admissible. Admissibility is enforced through the bounds-testing procedure of Pesaran et al. (2001), implemented via Natsiopoulos and Tzeremes (2024), which handles case-specific critical values internally. The admissible set is:

$$\mathcal{A}_{S1} \;=\; \{\, m \in \mathcal{L}_{S1} : F_{\text{bounds}}(m) \text{ rejects } H_0 \,\}. \tag{30}$$

Specifications that fail the bounds test—those for which the F-statistic falls below the lower critical value or into the inconclusive region—are excluded from further analysis. Among the survivors, model comparison proceeds not by scalar ranking but by mapping each specification into a two-dimensional fit–complexity plane:

$$m \;\mapsto\; \big(\, k(m),\; -2\log L(m) \,\big), \tag{31}$$

where $k(m)$ is the number of estimated parameters and $-2\log L(m)$ is twice the negative log-likelihood. In this representation, information criteria appear as linear iso-penalty contours—parallel lines whose slope is determined by the criterion's complexity penalty. Minimizing an information criterion selects the specification where a particular contour is tangent to the cloud of admissible models. Different criteria, with different slopes, select different tangency points.

[7] The step-dummy sets are nested: s_0 includes no dummies; $s_1 = \{d1974\}$; $s_2 = \{d1974, d1980\}$; $s_3 = \{d1956, d1974, d1980\}$. Each dummy takes the value 1 from its activation year onward. Their inclusion or exclusion is a researcher decision about the sample's event structure; the dummies produce level shifts in the capacity benchmark but do not alter the long-run slope $\hat{\theta}$, which remains a single scalar across the full sample in every specification.

Rather than selecting a single tangency point, S1 identifies the neighbourhood of specifications that occupy the low-IC region of the admissible cloud. The fattened frontier is defined as:

$$\mathcal{F}^{(0.20)} \;=\; \{\, m \in \mathcal{A}_{S1} : \mathrm{IC}(m) \le Q_{0.20}(\mathrm{IC}) \,\}, \tag{32}$$

the 20% of admissible specifications with the lowest information-criterion values—a credible specification band rather than a single winner. Five inferential objects are tracked across $\mathcal{F}^{(0.20)}$: the long-run parameter $\hat{\theta}$, the implied utilization series $\widehat{u}_t$, the capital-stock memory share s_K, the deterministic case c, and the step-dummy structure s. If these objects are stable across the fattened frontier, the capacity anchor is a property of the specification family. If they exhibit systematic variation—if the utilization series shifts level or the long-run multiplier changes sign as the specification moves along the frontier—the anchor is an artifact of winner selection.

Stage S1 confines the analysis to single-equation representations of the long-run restriction. The stability of the capacity anchor within the admissible ARDL class is a necessary but not sufficient condition for the identification claim: it remains possible that the anchor is a conditional artifact of the single-equation structure rather than a property of the data-generating process. Stage S2 asks whether the anchor survives when the long-run restriction is estimated as a system property rather than as a conditional single-equation relation.

4.5.1 S1 Results: IC Consensus

Both AIC and BIC select ARDL(1,3) under Cases 2–3, with and without Shaikh's dummies. The convergence is the strongest possible confinement of the specification axis. The lag structure is asymmetric—one lag on output, three on capital—and invariant to the dummy toggle. Output adjusts to shocks within a year; capital effects propagate over three. This asymmetry is physically coherent: a flow variable responds faster than a stock variable.

Shaikh's ARDL(2,4) ranks fourth by AIC under Cases 2–3, almost 3.5 AIC points behind the winner. The data does not want the second output lag or the fourth capital lag. The trend class (Cases 4–5) is dominated: its best specification loses to the no-trend winner by 2 AIC points.

Table 5: IC contest: Top-5 ARDL lag orders by model class (with impulse dummies)

Model class	PSS cases	p	q	AIC	BIC
No constant (Case 1)					
	Case 1	3	3	−260.72	−237.33
Constant (Cases 2–3)					
	Cases 2–3	**1**	**3**	**−270.23**	**−248.96**
	Cases 2–3	3	3	−270.15	−244.62
	Cases 2–3	2	3	−268.47	−245.07
	Cases 2–3	3	4	−266.81	—
Constant + trend (Cases 4–5)					
	Cases 4–5	1	3	−268.23	—

Global AIC and BIC winner bolded. Top entries per model class shown; full grid in Appendix. Shaikh's ARDL(2,4) = Cases 2–3 row 4.

ICOMP—the information complexity criterion of **?**, which penalizes not only the number of free parameters but the degree of interdependence among them through the maximal information complexity of the inverse Fisher information matrix—agrees with AIC and BIC at the same specification. Three criteria with fundamentally different penalty structures select the same neighbourhood. The specification axis is genuinely resolved.

Figure 3: S1 specification lattice. (a) Admissible cloud in the fit-complexity plane; Pareto frontier in dark red; Shaikh's ARDL(2,4) marked—dominated. (b) IC tangency points cluster at $k \approx 6$. (c) Informational domain $\hat{F}^{0.20}$: 15 specifications, $\theta \in [0.40, 0.71]$. (d) Utilization band from frontier specifications versus benchmarks.

4.5.2 S1 Results: Bounds Test Evidence

The PSS bounds test is evaluated at exact finite-sample critical values conditioned on $T = 65$, using $R = 40{,}000$ simulation draws. Asymptotic PSS critical values—the default in most software—are too permissive for this sample size; the finite-sample penalty makes the Case 2 rejection more conservative, and therefore more credible, than it would appear under asymptotic tables.

Table 6: PSS (2001) bounds test: dummy toggle comparison. ARDL(1,3), Cases I–V.

	With dummies		Without dummies	
Case	F	p	F	p
Case 1	2.03	0.272	5.19	0.023
Case 2	**6.22**	**0.009**	**7.03**	**0.004**
Case 3	6.44	0.038	5.27	0.077
Case 4	4.22	0.145	3.73	0.213
Case 5	2.41	0.693	1.32	0.898

Exact p-values, $T = 65$, $R = 40{,}000$ draws. Removing dummies concentrates evidence into Case 2 alone; Case 3 weakens from $p = 0.038$ to $p = 0.077$. t-bounds test structurally undefined for Cases 2 and 4.

Cointegration is found under Case 2 ($p = 0.004$) and marginally under Case 3 ($p = 0.077$). Every specification that includes a trend (Cases 4–5) fails. The trend exclusion is now established through three independent channels: the IC contest, the bounds test, and the S0 direct estimation. All three converge.

The t-bounds test never rejects across any case or dummy toggle. The system cointegrates in the joint F sense but the speed at which output adjusts toward equilibrium is indistinguishable from zero. The capacity utilization series implied by the ARDL is constructed from a cointegrating residual whose error-correction dynamics cannot be distinguished from zero—the "equilibrium" to which the series is referenced barely functions as an attractor within the sample.

4.5.3 S1 Results: What Case 2 Means

Case 2 restricts the intercept to the long-run equation: $\ln Y^* = \mu + \theta \ln K$. No drift in the short-run dynamics. The system has one fixed equilibrium level across the entire sample—one μ, one θ, 1947–2011.

This restriction is the necessary condition for Shaikh's demeaning normalization to be internally coherent. The transformation $\hat{u}_t = \exp(\mathrm{ECT}_t - \overline{\mathrm{ECT}})$ requires that the ECT have a stable mean across the sample. Under Case 2 (no short-run drift), it does. Under Case 3 (drift permitted), the mean migrates and the demeaning produces a moving target disguised as a fixed normalization. The data indicates cointegration strongest precisely where the normalization is valid.

But "one fixed equilibrium across 1947–2011" means the accumulation regime did not shift—across the Fordist golden age, the stagflationary crisis, the neoliberal turn, financialization, and the Great Recession. The confinement that makes the cointegrating evidence strongest is

the same confinement that most aggressively suppresses the regime-dependence the theoretical framework predicts.

4.5.4 What S1 Resolves and What It Leaves Open

S1 confines the specification axis completely. The IC-preferred specification is ARDL(1,3) under Cases 2–3. AIC, BIC, and ICOMP converge. The lag structure is stable. The trend is excluded by three independent channels.

The S0 pathologies persist under proper specification search. The persistence amplification is not a lag-order artifact—it is a structural property of the ARDL long-run multiplier near the unit root boundary. The weak error correction is not fixable by better specification—the t-test fails under the IC-preferred order just as it did under Shaikh's arbitrary (2,4). The method works—cointegration exists—but the ceiling is visible.

The ARDL framework pre-commits to three architectural constraints that define this ceiling: rank = 1 (assumed, not tested), normalization on output (capital treated as weakly exogenous), and long-run multiplier recovery through the near-zero denominator (the persistence amplification mechanism). No specification search can release these constraints. They are properties of the single-equation framework, not of any particular specification within it.

S2 releases all three simultaneously by moving to the Johansen VECM—rank determination replaces the assumed rank, the loading matrix reveals which variables adjust, and the eigenvector estimator recovers θ without passing through a near-zero divisor.

Figure 4: S0–S1 closing figure: capacity utilization specification fan. 24 series from the full {pairing × lag order × trend × dummy} grid, faceted by capital stock concept. Left: Gross K (blue)—tight band. Right: Net K (orange)—wide band, ψ_t contamination visible. Benchmarks: Shaikh u_K (dark red), FRB (black).

4.6 Stage S2: System-Level VECM Replication

Stage S1 established that the capacity anchor is stable within the admissible ARDL specification class: the long-run multiplier and the implied utilization series do not depend on which information criterion selects the winning specification, provided the specification lies within the credible specification band $\mathcal{F}^{(0.20)}$. This is a necessary but not sufficient condition for the identification claim. The ARDL framework estimates the long-run restriction as a conditional single-equation relation, treating capital as the forcing variable and recovering the cointegrating vector from the conditional error-correction form. It remains possible that the capacity anchor is an artifact of this conditioning structure—that the long-run restriction would not survive if output and capital were allowed to adjust simultaneously to disequilibrium. S2 closes this sufficiency gap by re-estimating the long-run restriction as a system property.

The theoretical object under replication is the same as in S1. In Shaikh's framework, the log-level relation between output and capital decomposes as:

$$\ln Y_t \;-\; \theta \ln K_t \;=\; \ln u_t, \tag{33}$$

where θ is the long-run output–capital multiplier and u_t is capacity utilization. Cointegration between $\ln Y$ and $\ln K$ identifies the capacity transformation relation; the cointegrating residual represents utilization dynamics; and the maintained closure $u = 1$ in the long run fixes the theoretical anchor. S2 holds fixed the same element that S1 held fixed: the closure $u = 1$ and

the interpretation of the cointegrating residual as capacity utilization. What S2 interrogates is whether the cointegrating vector β and the implied utilization series $\widehat{u}_t$ are properties of the joint system or conditional artifacts of the single-equation structure. The shift from S1 to S2 is a shift in the identification environment, not in the theoretical object.

The question carries the same geometric framing as S1, extended to system estimation. The null hypothesis is that the capacity anchor recovered in S1 is a conditional artifact of single-equation estimation—that the long-run restriction is not identified as a system property when output and capital are permitted to adjust jointly to the error-correction term. The alternative is that the cointegrating vector β and the implied utilization series are stable system properties, recoverable across the admissible VECM specification space. As in S1, the argument is made geometrically rather than by formal test. Information criteria remain coordinates in the fit–complexity plane, not parametric penalty functions: each criterion selects a different point in the admissible cloud, and the question is whether the capacity anchor varies across the credible specification band or is invariant to the choice of coordinate system.

The specification space is again organized as a lattice, now extended to accommodate the additional dimensions that system estimation introduces. Each candidate model $m \in \mathcal{L}_{S2}$ is indexed by a six-dimensional tuple:

$$\mathcal{L}_{S2} \;=\; \{\, (m,\, r,\, p,\, q,\, d,\, h) \,\}, \tag{34}$$

where m is the system dimension, r is the cointegration rank, p is the lag depth governing short-run dynamics, q indexes the short-run memory architecture distributing lagged adjustment across variables, d indexes the deterministic branch, and h indexes the step-dummy structure. Each specification implies a reduced-rank vector error-correction representation:

$$\Delta X_t \;=\; \alpha\beta' X_{t-1} \;+\; \sum_{i=1}^{p-1} \Gamma_i\, \Delta X_{t-i} \;+\; \det(d, h) \;+\; \varepsilon_t, \tag{35}$$

where $\Pi = \alpha\beta'$ with $\operatorname{rank}(\Pi) = r$, estimated via maximum likelihood (Stigler, 2020). Two system sizes are economically motivated. The $m = 2$ system sets $X_t = (\ln Y_t,\, \ln K_t)'$ and imposes $r = 1$, reproducing Shaikh's core bivariate structure at the system level: one cointegrating vector identifies the capacity transformation relation. The $m = 3$ system extends the state vector to $X_t = (\ln Y_t,\, \ln K_t,\, \ln e_t)'$, where $\ln e_t$ is the log rate of exploitation, and admits $r = 2$: a second cointegrating vector opens the possibility of identifying reserve-army dynamics alongside the capacity relation. The $m = 3$ system requires a third variable: Shaikh's corrected corporate profit share, Profshcorp_t, computed as the ratio of corrected corporate net operating surplus to corrected corporate value added, serves as the empirical proxy for the rate of exploitation. The correction applies the same imputed-interest adjustment used in constructing GVAcorp_t; exact source-table mapping and line-item formulas are provided in Appendix B. The deterministic branch d governs where constant and trend terms enter the system: d_0 imposes no deterministic term; d_1 restricts the constant to the long-run relation only; d_2 permits an unrestricted short-run mean; d_3 admits both long-run and short-run deterministic components. The step-dummy structure h enters the system as exogenous regressors in the short-run dynamics, producing level shifts in the deterministic component without altering the cointegrating slope θ, which remains a single scalar across the full sample: h_0 imposes no dummies; h_1 includes a step dummy at 1974; h_2 includes step dummies at 1956, 1974, and 1980.[8]

Admissibility in the VECM environment is enforced through a triple gate rather than through a single bounds test. A specification is admissible if and only if estimation converges without

[8]The step dummies are defined identically to S1: $d\text{YYYY} = 1$ if year $\geq$ YYYY. Their interpretation as historical episodes—and the question of whether they mark regime boundaries at which θ itself changes—is deferred to S3, where parametric change is tested within the VECM framework. In S0–S2, the dummies are specification choices that shift the level of the capacity benchmark, not slope-change diagnostics.

degeneracy, the imposed cointegration rank is consistent with trace and maximum-eigenvalue test statistics (Johansen, 1991), and the companion matrix exhibits exactly $m - r$ unit roots with no explosive eigenvalues. The admissible set is:

$$\mathcal{A}_{S2} \;=\; \{\, m \in \mathcal{L}_{S2} : \text{convergence} \;\wedge\; \text{rank satisfied} \;\wedge\; \text{stability satisfied} \,\}. \tag{36}$$

Among the admissible specifications, model comparison proceeds through the same fit–complexity mapping as in S1:

$$m \;\mapsto\; \big(\, k(m),\; -2\log L(m) \,\big), \tag{37}$$

where $k(m)$ now counts all estimated parameters in the system— cointegrating vectors, adjustment coefficients, short-run dynamics, and deterministic terms. The fit–complexity plane is identical in construction to that of S1; what changes is the admissibility gate and the dimensionality of the specifications populating the cloud.

Rather than selecting a single tangency point, S2 identifies the informational domain Ω_{20}—the 20% of admissible specifications achieving the lowest log-likelihood loss:

$$\Omega_{20} \;=\; \{\, m \in \mathcal{A}_{S2} : -2\log L(m) \le Q_{0.20}(-2\log L) \,\}. \tag{38}$$

This is the system-level counterpart of S1's fattened frontier $\mathcal{F}^{(0.20)}$: a credible specification band, not a single winner. Six inferential objects are tracked across Ω_{20}: the cointegrating vector β, governing the long-run capacity transformation relation; the implied utilization series $\widehat{u}_t$; the adjustment speed matrix α, which determines which variables bear the error-correction burden; the capital-side short-run memory share s_K; the deterministic closure structure d; and the step-dummy structure h. If these objects are stable across Ω_{20}—if the cointegrating vector does not rotate, the utilization series does not shift level, and the error-correction burden does not migrate across equations as the specification moves through the informational domain—the capacity anchor is a system property. If they exhibit systematic variation, the anchor recovered in S1 was a conditional artifact that does not survive system identification.

A positive S2 result licenses a specific claim: the capacity anchor is not a conditional artifact of single-equation ARDL estimation but a property of the joint data-generating process, recoverable across the admissible VECM specification space. The $m = 3$ result opens a further possibility. If a second cointegrating vector emerges when the rate of exploitation enters the state vector, the system admits reserve-army dynamics alongside the capacity transformation relation—and identifying both relations, including their interaction, requires structural restrictions on the cointegrating space and the adjustment matrix that the grid search of S2 leaves unrestricted. That identification exceeds the bivariate system's capacity.

The need for structural completion can be stated precisely. The informational domain Ω_{20} delivers a family of admissible VECM specifications whose likelihood surface defines candidate cointegrating vectors recovered from the canonical correlation decomposition—reduced-form relations that are not yet structurally identified. Separating the capacity transformation relation from the reserve-army relation requires a rotation of the canonical correlation space, disciplined by the theoretical prior:

$$\hat{e}_t \;=\; \mu \;+\; \lambda\,\hat{u}_t \;+\; \sum_h \kappa_h\, DM_{h,t}, \qquad \lambda < 0, \tag{39}$$

where the sign restriction $\lambda < 0$ encodes the invariant that rising capacity utilization exerts a negative cyclical effect on exploitation pressure, transmitted via the employment rate which need not enter the state vector explicitly. This is not a Johansen restriction and not a reduced-rank restriction; it is the economic prior that identifies the rotation. Completing that rotation requires imposing theoretically motivated restrictions on β and α jointly—restrictions the unrestricted VECM grid of S2 does not supply.

Through S0–S2, the long-run parameter θ is a single scalar across the full sample in every specification. The question of whether θ varies across accumulation regimes—whether the fixed parameter suppresses structural heterogeneity of the kind the accounting laws of §3.2 predict—is the object of Stage S3, where parametric change is tested within the VECM framework through historical toggles under a regulationist interpretation of the Fordist/post-Fordist boundary.

4.6.1 Results

Table 7 reports the IC-selected VECM specifications. In the bivariate system, all four criteria select lag depth $p = 1$ with $\hat{\theta}_{S2} \in [0.870, 0.900]$; the Ω_{20} mean is $\hat{\theta} = 0.803$. Table 8 summarizes the funnel: of 48 candidates, 21 pass the triple gate (43.8%).

The trivariate system produces a sign split: AIC and HQ recover $\hat{\theta} = -1.20$ under the unrestricted-constant specification (d_3), while BIC and ICOMP recover $\hat{\theta} \approx 0.93$ under d_0. The sign degeneracy does not disqualify the trivariate system — it signals that the reduced-form canonical correlations have not been rotated into economically identified vectors. Whether the rotation (eq. 39) resolves the degeneracy is tested alongside the regime-break analysis, where the trivariate system re-enters under structural identification constraints.

Table 7: S2 IC winners: bivariate and trivariate VECM

m	IC	p	Det.	Dummies	$\hat{\theta}$	$-2\log L$
2	AIC	1	d_2	h_1	0.885	−572.5
2	BIC	1	d_0	h_0	0.870	−560.4
2	HQ	1	d_2	h_1	0.885	−572.5
2	ICOMP	1	d_0	h_1	0.900	−566.5
3	AIC	3	d_3	h_0	−1.195	−802.3
3	BIC	3	d_0	h_0	0.930	−783.9
3	HQ	3	d_3	h_0	−1.195	−802.3
3	ICOMP	3	d_0	h_0	0.930	−783.9

Bivariate Ω_{20} mean: $\hat{\theta} = 0.803$. Trivariate sign split: reduced-form degeneracy, rotation pending.

Table 8: S2 VECM lattice summary

m	Stage	N	Description
2	Full lattice	48	(p, d, h) grid
2	Admissible	21	Johansen rank ≥ 1 (43.8%)
2	Ω_{20}	5	Lowest 20% of $-2\log L$
2	IC band	[0.87, 0.90]	All 4 ICs
3	Admissible	low	∼2% admissibility
3	Sign split	2/4	AIC/HQ: $\hat{\theta} < 0$; BIC/ICOMP: $\hat{\theta} > 0$

Both estimators indicate $\hat{\theta} < 1$: over-mechanization is robust across the single-equation and system frameworks. The point estimates do not coincide: $\hat{\theta}_{S1} \approx 0.57$ versus $\hat{\theta}_{S2} \approx 0.88$. Interpreted through the measurement framework, this gap is the empirical counterpart of the denominator fragility diagnosed in the four-pairing exercise (Table 4): the ARDL recovers $\hat{\theta}$ through $\sum \hat{\varphi}_j/(1 - \sum \hat{\psi}_i)$, where the divisor is approximately 0.12; the VECM eigenvector estimator bypasses that divisor. The level discrepancy is consistent with method-induced depression of the ARDL point estimate, not with a structural difference in the cointegrating relationship.

The trivariate sign split indicates that rank 1 is the identification boundary of the unrestricted bivariate system. This conclusion is conditional on treating the unrestricted VECM as the identification environment; it is possible that the degeneracy reflects entanglement of the capacity and reserve-army relations that the sign restriction $\lambda < 0$ can resolve. Two questions

pass forward: whether $\hat{\theta}$ is stable across the Fordist/Post-Fordist periodization, and whether the trivariate degeneracy resolves under structural rotation.

4.7 Cross-Stage Summary

Table 9 collects the $\hat{\theta}$ trajectory. The qualitative invariant — $\hat{\theta} < 1$, consistent with over-mechanization — holds across every admissible specification in every stage. The quantitative variation is governed by two identified sources: the denominator regime (gross vs. net capital stock, diagnosed in S0) and the estimator class (ARDL near-singular divisor vs. VECM eigenvector, diagnosed in S2). Neither source indicates a different economic relationship; both are properties of the parametric recovery of the long-run multiplier at this empirical site.

Table 9: Cross-stage $\hat{\theta}$ trajectory

Stage	Estimator	$\hat{\theta}$	Note
S0	ARDL(2,4), Case II	0.745	Sole passing case
S0	ARDL(2,4), Cases IV–V	0.541	Trend collapse (−27%)
S0	Four pairings, gross K	[0.54, 0.75]	GVA/NVA < 1 pp
S0	Four pairings, net K	[0.11, 0.81]	Denominator-amplified
S1	ARDL(1,3), Case II	0.565	4/5 ICs; no dummies
S1	RICOMP dissenter	0.397	Sandwich penalty
S1	$\mathcal{F}^{(0.20)}$ frontier	[0.40, 0.71]	15 specs
S2	VECM $m=2$, IC band	[0.87, 0.90]	Eigenvector
S2	VECM $m=2$, Ω_{20}	0.803	Domain mean
S2	VECM $m=3$	sign split	Rotation pending

All stages: $\hat{\theta} < 1$ (over-mechanization).
S1→S2 gap: denominator fragility (ARDL divisor vs. VECM eigenvector).
$m = 3$ sign degeneracy: structural rotation deferred.

The confined bivariate specification ($m = 2$, $p = 1$, Case II) and the unresolved trivariate degeneracy pass forward to the regime-break analysis, which tests whether $\hat{\theta}$ is stable across the Fordist/Post-Fordist periodization and whether the sign restriction $\lambda < 0$ resolves the trivariate entanglement under structural identification constraints.

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A Derivation of the Accumulation Dynamics under Unbalanced Growth

This appendix provides the full derivation of the dynamical equations summarized in §3.2.4 and §3.2.5. All results follow from three ingredients: the accounting identity $\hat{k} + \delta = \phi^p B$ (equation 1), the unbalanced growth closure $\hat{y}^p = \theta(\Lambda)\hat{k}$ (equation 3), and the investment-savings closure $\phi^p = s$ (equation 7).

A.1 The Pure Closure: Derivation of the Bernoulli ODE

Begin from the accounting identity under the investment-savings closure:

$$\hat{k} + \delta = s\,B \tag{40}$$

Since s is constant, differentiate both sides with respect to time:

$$\frac{d\hat{k}}{dt} = s\,\dot{B} \tag{41}$$

Rewrite $\dot{B}$ in terms of the growth rate of capital productivity:

$$\dot{B} = B\,\hat{b} \tag{42}$$

Substituting into equation (41) and using $sB = \hat{k} + \delta$ from the accounting identity:

$$\frac{d\hat{k}}{dt} = s\,B\,\hat{b} = (\hat{k} + \delta)\,\hat{b} \tag{43}$$

This is the acceleration identity: the rate of change of the net accumulation rate equals the gross accumulation rate times the growth rate of capital productivity. It holds for any closure on $\hat{b}$.

Now substitute the unbalanced growth closure. From equation (4):

$$\hat{b} = (\theta(\Lambda) - 1)\,\hat{k} \tag{44}$$

Inserting into equation (43):

$$\boxed{\frac{d\hat{k}}{dt} = (\theta(\Lambda) - 1)\,\hat{k}\,(\hat{k} + \delta)} \tag{45}$$

This is a Bernoulli equation of order 2 in $\hat{k}$. The nonlinearity arises from the product $\hat{k}(\hat{k} + \delta)$: the acceleration of accumulation depends on both the current net rate and the gross rate, coupled through the regime parameter.

A.2 Analytical Solution via the Bernoulli Substitution

Expand the right-hand side of equation (45):

$$\frac{d\hat{k}}{dt} = (\theta - 1)\,\hat{k}^2 + (\theta - 1)\,\delta\,\hat{k} \tag{46}$$

This is a Bernoulli equation of the form $\dot{x} = P(t)\,x + Q(t)\,x^n$ with $n = 2$, $P = (\theta - 1)\delta$, and $Q = (\theta - 1)$, all constant. Apply the standard substitution $v \equiv \hat{k}^{1-n} = \hat{k}^{-1}$:

$$\dot{v} = -\hat{k}^{-2}\,\dot{\hat{k}} \tag{47}$$

Dividing equation (46) by $\hat{k}^2$:

$$\hat{k}^{-2}\,\frac{d\hat{k}}{dt} = (\theta - 1) + (\theta - 1)\,\delta\,\hat{k}^{-1} \tag{48}$$

Substituting $v = \hat{k}^{-1}$ and $\dot{v} = -\hat{k}^{-2}\dot{\hat{k}}$:

$$\frac{dv}{dt} = -(\theta - 1)\,\delta\,v - (\theta - 1) \tag{49}$$

This is a first-order linear ODE in v with constant coefficients. Let $\alpha \equiv (\theta - 1)\delta$. The equation is:

$$\dot{v} + \alpha\,v = -(\theta - 1) \tag{50}$$

The integrating factor is $e^{\alpha t}$. The general solution is:

$$v(t) = \left(v_0 + \frac{\theta - 1}{\alpha}\right) e^{-\alpha t} - \frac{\theta - 1}{\alpha} \tag{51}$$

Since $\alpha = (\theta - 1)\delta$, the ratio simplifies: $(\theta - 1)/\alpha = 1/\delta$. Therefore:

$$v(t) = \left(v_0 + \frac{1}{\delta}\right) e^{-(\theta-1)\delta\,t} - \frac{1}{\delta} \tag{52}$$

Substituting back $v = 1/\hat{k}$:

$$\boxed{\hat{k}(t) = \frac{1}{\left(\frac{1}{\hat{k}_0} + \frac{1}{\delta}\right) e^{-(\theta-1)\delta\,t} - \frac{1}{\delta}}} \tag{53}$$

where $\hat{k}_0 = \hat{k}(0)$ is the initial net accumulation rate.

A.3 Equilibria and Stability Analysis

The equilibria of equation (45) are the values $\hat{k}^*$ such that $d\hat{k}/dt = 0$. Factoring:

$$(\theta - 1)\,\hat{k}\,(\hat{k} + \delta) = 0 \tag{54}$$

For $\theta \neq 1$, the roots are:

$$\hat{k}_1^* = 0, \qquad \hat{k}_2^* = -\delta \tag{55}$$

Interpretation. $\hat{k}^* = 0$ means the net capital stock is stationary: gross investment exactly covers depreciation ($I/K = \delta$). $\hat{k}^* = -\delta$ means gross investment is zero: the capital stock depreciates without replacement.

Stability. Linearize around each equilibrium. The Jacobian of $f(\hat{k}) = (\theta - 1)\hat{k}(\hat{k} + \delta)$ is:

$$f'(\hat{k}) = (\theta - 1)(2\hat{k} + \delta) \tag{56}$$

At $\hat{k}_1^* = 0$:

$$f'(0) = (\theta - 1)\,\delta \tag{57}$$

- Over-mechanization ($\theta < 1$): $f'(0) < 0$ — the stagnation equilibrium is **locally stable**.
- Under-mechanization ($\theta > 1$): $f'(0) > 0$ — the stagnation equilibrium is **unstable**.

At $\hat{k}_2^* = -\delta$:

$$f'(-\delta) = -(\theta - 1)\,\delta \tag{58}$$

- Over-mechanization ($\theta < 1$): $f'(-\delta) > 0$ — the zero-investment equilibrium is **unstable**.
- Under-mechanization ($\theta > 1$): $f'(-\delta) < 0$ — the zero-investment equilibrium is **locally stable**.

Summary of the phase portrait:

Regime	$\hat{k}^* = 0$	$\hat{k}^* = -\delta$	**In the feasible region** $\hat{k} > 0$
Over-mechanization ($\theta < 1$)	Stable	Unstable	$\hat{k} \to 0$: accumulation stagnates
Under-mechanization ($\theta > 1$)	Unstable	Stable	$\hat{k} \to \infty$: accumulation explodes

Table 10: Stability of equilibria under the pure unbalanced growth closure.

Under over-mechanization, any positive $\hat{k}_0 > 0$ decays to zero: the economy converges to a state where net accumulation vanishes. Under under-mechanization, any positive $\hat{k}_0 > 0$ diverges: the economy accelerates without bound. In neither case does a feasible interior attractor exist. This confirms the regime classification in Table 1 and the instability result stated in §3.2.4.

Verification from the explicit solution. In equation (53), the denominator contains the term $e^{-(\theta-1)\delta t}$. Under over-mechanization ($\theta < 1$), the exponent is positive and the exponential grows, driving the denominator toward $+\infty$ and $\hat{k}(t) \to 0$. Under under-mechanization ($\theta > 1$), the exponent is negative and the exponential decays, driving the denominator toward $-1/\delta < 0$ and $\hat{k}(t) \to -\delta$ — but since $\hat{k}$ passes through zero or diverges before reaching $-\delta$, the solution blows up in finite time. Both regimes are consistent with the linearized stability analysis.

A.4 The Extended Closure: Derivation of the Quadratic ODE

Under Shaikh's extended closure $\hat{y}^p = \theta\hat{k} + \beta$, the growth rate of capital productivity becomes:

$$\hat{b} = \hat{y}^p - \hat{k} = (\theta - 1)\hat{k} + \beta \tag{59}$$

Substituting into the acceleration identity (equation 43):

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta)\left[(\theta - 1)\hat{k} + \beta\right] \tag{60}$$

Expanding:

$$\frac{d\hat{k}}{dt} = (\theta - 1)\hat{k}^2 + [(\theta - 1)\delta + \beta]\,\hat{k} + \beta\delta \tag{61}$$

The equilibria are the roots of $(\hat{k} + \delta)[(\theta - 1)\hat{k} + \beta] = 0$:

$$\hat{k}_1^* = -\delta, \qquad \hat{k}_2^* = \frac{-\beta}{\theta - 1} = \frac{\beta}{1 - \theta} \tag{62}$$

Stability of the new equilibrium. The Jacobian of $g(\hat{k}) = (\hat{k} + \delta)[(\theta - 1)\hat{k} + \beta]$ is:

$$g'(\hat{k}) = 2(\theta - 1)\hat{k} + (\theta - 1)\delta + \beta \tag{63}$$

At $\hat{k}_2^* = \beta/(1 - \theta)$:

$$g'\left(\frac{\beta}{1 - \theta}\right) = \frac{-2\beta(\theta - 1)}{1 - \theta} + (\theta - 1)\delta + \beta = 2\beta + (\theta - 1)\delta + \beta = 3\beta + (\theta - 1)\delta \tag{64}$$

Under over-mechanization ($\theta < 1$) with $\beta > 0$: the sign depends on the balance between 3β and $(1-\theta)\delta$. For the empirically relevant case where β is small relative to δ (secular technical progress is slower than the depreciation rate), $g'(\hat{k}_2^*) < 0$ and the equilibrium is **locally stable**. This confirms that autonomous technical progress provides the interior attractor that the pure closure lacks.

Limiting behavior. As $\beta \to 0$: $\hat{k}_2^* \to 0$, the new equilibrium collapses onto the stagnation point $\hat{k}_1^* = 0$ of the pure closure, the quadratic degenerates to $(\theta-1)\hat{k}(\hat{k}+\delta) = 0$, and the Bernoulli ODE is recovered. The pure closure is the $\beta = 0$ limit of Shaikh's extended specification.

A.5 Vieta's Formulas for the Quadratic Equilibrium Structure

Writing the equilibrium condition $(\theta-1)\hat{k}^2 + [(\theta-1)\delta+\beta]\hat{k} + \beta\delta = 0$ in standard form $a\hat{k}^2 + b\hat{k} + c = 0$ with:

$$a = \theta - 1, \qquad b = (\theta-1)\delta + \beta, \qquad c = \beta\delta \tag{65}$$

Vieta's formulas give:

$$\hat{k}_1^* + \hat{k}_2^* = -\frac{b}{a} = -\delta - \frac{\beta}{\theta-1} \tag{66}$$

$$\hat{k}_1^* \cdot \hat{k}_2^* = \frac{c}{a} = \frac{\beta\delta}{\theta-1} \tag{67}$$

Product of roots. The sign of the product depends on $(\theta-1)$:

- Over-mechanization ($\theta < 1$): $\hat{k}_1^* \cdot \hat{k}_2^* < 0$ — one positive root ($\hat{k}_2^* > 0$) and one negative ($\hat{k}_1^* = -\delta$). A feasible attractor exists.
- Under-mechanization ($\theta > 1$): $\hat{k}_1^* \cdot \hat{k}_2^* > 0$ — both roots negative. No feasible positive equilibrium.

Sum of roots. The depreciation rate δ anchors the sum. The term $\beta/(\theta-1)$ scales autonomous progress by the inverse degree of unbalanced growth: as $\theta \to 1$ from either side, this term diverges, reflecting the amplification of the structural accumulation imbalance near the knife-edge.

Degeneracy. As $\theta \to 1$, $a \to 0$ and the quadratic degenerates. Simultaneously, $\hat{k}_2^* = \beta/(1-\theta) \to \pm\infty$. The joint condition $\theta = 1$ and $\beta = 0$ defines the degeneracy locus $\mathcal{D}$ at which the system has no determinate growth rate.

B Data Glossary and Construction Ledger

B.1 Series Glossary

B.2 Imputed-Interest Correction

The NIPA corporate value added figure includes imputed financial intermediation services as a cost to non-financial corporations. The corporate imputed-interest adjustment removes this component:

$$\text{CorpImpIntAdj}_t \equiv -\,\text{BankMonIntPaid}_t - \text{CorpNFNetImpIntPaid}_t, \tag{68}$$

Table 11: Replication series: symbols, sources, and transformations

Symbol / Code	Variable	Source (table, line)	Transformation
GVAcorp_t	Corporate gross value added (corrected)	NIPA 1.14 (L1) + imputed-interest adj. (Table 7.11)	nominal
KGCcorp_t	Corporate gross capital stock (GPIM-adjusted)	Fixed Asset 6.1 (L9); IRS Statistics of Income; GPIM recursion	nominal
KNCcorp_t	Corporate net capital stock (GPIM-adjusted)	same as above	nominal
KNCcorpbea_t	Corporate net capital stock (BEA 2011 official)	Fixed Asset 6.1 (L9)	nominal
p_t	Common deflator	NIPA 1.1.9 (L1)	index
y_t	Real log output	constructed	$\ln(\text{GVAcorp}_t/p_t)$
k_t	Real log capital	constructed	$\ln(\text{KGCcorp}_t/p_t)$
Profshcorp_t	Corporate profit share (corrected)	NIPA 1.14 (L8, L1−L2) + imputed-interest adj.	ratio

Note: All BEA and NIPA data downloaded in the replication vintage. Shaikh's original series used data downloaded in 2011 (Shaikh, 2016, Appendix 6.7, fn. 1); discrepancies attributable to subsequent BEA revisions are possible and noted where material.

Source: Real Economic Analysis Data Tables

where BankMonIntPaid_t is NIPA Table 7.11 (lines 4+44+73 minus lines 28+52+91) and CorpNFNetImpIntPaid is NIPA Table 7.11 (line 74 minus line 53). In 2009 the adjustment raises corporate net operating surplus by approximately 10 %.

B.3 Corrected Corporate Output: GVAcorp_t

$$\text{GVAcorp}_t \;\equiv\; \text{GVAcorpnipa}_t \;+\; \text{CorpImpIntAdj}_t, \tag{69}$$

where GVAcorpnipa_t is NIPA Table 1.14 (line 1) and CorpImpIntAdj_t is defined in equation (68).

B.4 Reconstructed Corporate Capital Stock: KGCcorp_t

GPIM accumulation rule. The current-cost capital stock is accumulated as:

$$K_t \;=\; IG_t \;+\; z_t^* \cdot K_{t-1}, \qquad z_t^* \;\equiv\; (1 - z_t)\,\frac{p_t^K}{p_{t-1}^K}, \tag{70}$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* is the survival-and-revaluation factor that reflates the surviving stock to current replacement cost.

Three adjustments (applied before the recursion).

1. **Depletion rates.** BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption, implying higher depreciation and lower retirement rates consistent with physical exhaustion.

2. **Initial values.** BEA 1993 initial values anchored to 1925 replace the BEA 2011 starting values. The 1993 values are lower, producing faster accumulation growth and convergence to the official path by approximately 1977.

3. **Great Depression correction.** The IRS corporate book value index (**?**, Series V 115, pp. 924–926) is applied over 1925–1947, rescaling the BEA historical stock by the IRS/BEA ratio each year. The GPIM recursion resumes from 1947. Net effect: KNCcorp_t is approximately 28 % lower in 1947 than the BEA 2011 official measure.

Inventories. IRS corporate balance sheet inventory ratios are extrapolated over 1947–1989 via a linear fit to the 1990–2011 IRS/BEA ratio, rescaled to the adjusted GPIM stocks, and added to KGCcorp_t. Source: IRS Statistics of Income 1926–2011; FRB Flow of Funds FL105015205.A for non-financial corporate inventories in current cost.

Accuracy. The average ratio of the GPIM approximation to the BEA current-cost stock over 1947–2005 is 99.60 %.

Full recursion inputs and depletion-rate construction are in Shaikh (2016, Appendix 6.7, Section V; Appendix Table 6.8.II.3).

B.5 Common Deflator: p_t

The replication uses the GDP implicit price deflator (BEA NIPA Table 1.1.9, line 1) as p_t. This is a workaround: Shaikh's Appendix 6.8 tables do not tabulate the output-side deflator used in the corporate utilization exercise (see §**??**). The transformed series are:

$$y_t \equiv \ln\left(\frac{\text{GVAcorp}_t}{p_t}\right), \qquad k_t \equiv \ln\left(\frac{\text{KGCcorp}_t}{p_t}\right). \tag{71}$$

B.6 Corrected Corporate Profit Share: Profshcorp_t

The same imputed-interest adjustment applied to GVAcorp_t is applied to both the numerator and denominator of the profit share:

$$\text{NOScorp}_t \equiv \text{NOScorpnipa}_t + \text{CorpImpIntAdj}_t, \tag{72}$$

$$\text{VAcorp}_t \equiv \text{VAcorpnipa}_t + \text{CorpImpIntAdj}_t, \tag{73}$$

$$\text{Profshcorp}_t \equiv \frac{\text{NOScorp}_t}{\text{VAcorp}_t}, \tag{74}$$

where NOScorpnipa_t is NIPA Table 1.14 (line 8) and VAcorpnipa_t is NIPA Table 1.14 (line 1 minus line 2). Profshcorp_t is constructed in the S0 data environment but does not enter the S0 or S1 identification objects; its first use is as the empirical proxy for $\ln e_t$ in the S2 $m = 3$ VECM system.

C Electric Motor Utilization: Physical Measures Lineage

The Electric Motor Utilization index represents the founding device of the physical-intensity tradition in capacity measurement. Unlike survey-based approaches, which elicit managerial reports of operating rates relative to a self-defined normal, the EMU methodology grounds the utilization ratio in an engineering relationship between installed motor capacity and observed electricity consumption. Table 12 documents the methodology, data sources, and successive extensions of this lineage from Foss (1963) through Shaikh (1992, 2016) to Nikiforos (2016), who formalized the physical-intensity critique as a micro-founded theoretical result and proposed the Average Workweek of Capital as the contemporary analogue.

Table 12: Electric Motor Utilization (EMU) as a Proxy for Capacity Utilization: Physical Measures Lineage

Feature	Description
Time Span	1929–1954 (manufacturing); extended to 1899–1963 by Shaikh (1992).
Data & Methodology	Calculates potential motor output (in kWh) from horsepower of installed electric motors × 8,760 hours/year × 0.746 kWh per hp-hour ÷ 0.9 (efficiency adjustment). Actual electricity consumption is then compared to this theoretical maximum. Primary sources: U.S. Federal Power Commission (1946); US Bureau of Census (1966, 1967); U.S. Bureau of the Census (1975); US Department of Commerce (1986).
Main Results	Substantial long-term increases in motor utilization from 1929 to 1954: utilization rates rose from 15.9% (1929) to 20.9% (1954), indicating rising intensity in capital use even as labour hours declined (Foss, 1963).
Critique of FRB	Indirect. Suggests that survey-based measures overlook mechanical intensity and changes in input use; implicitly argues for a more objective, physically grounded metric capable of registering greater long-run variation than survey-elicited operating rates.
Extensions by Shaikh (1992)	Adapts and extends the EMU method using Census data and Christensen and Jorgenson (1969) to build a continuous series from 1899–1963, adjusting pre-1939 estimates and interpolating from industry-level motor data. Spliced with McGraw-Hill survey data to produce a utilization measure covering 1947–1985. The extended series diverges from FRB in amplitude, trend, and cyclical pattern, capturing episodes the FRB series suppresses (Shaikh, 2016, Appendix 6.6).
Formalization by Nikiforos (2016)	Re-introduces the Average Workweek of Capital (AWW) — hours per week that capital equipment is operated — as the contemporary physical-intensity analogue to the EMU ratio. Formalizes the critique of survey-based measures through the plant-manager thought experiment: because adding or dropping shifts changes the normal against which managers report, the survey denominator moves with the numerator. Stationarity of the FRB series is thereby a construction artefact, not a property of the underlying utilization dynamics. AWW-based estimates exhibit non-stationary behaviour consistent with the EMU lineage.
Primary Sources	U.S. Federal Power Commission (1946); US Bureau of Census (1966); US Bureau of Census (1967); U.S. Bureau of the Census (1975); US Department of Commerce (1986); Christensen and Jorgenson (1969); Foss (1963); Shaikh (1992, 2016); Nikiforos (2016).

Source: Own elaboration based on Foss (1963), Shaikh (1992, 2016), and Nikiforos (2016).